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Department of Artificial Intelligence (AI) and Data Science

Major Project Report

A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Artificial Intelligence (AI) and Data Science

by

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Academic Year 2026–2027

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Approval Sheet

This Project Report entitled "A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains" submitted by "Doctor UmerKhan JunaidKhan", "Kerkar Suyash Satish", "Esaf Shanum Shakil", "Patil Rutuja Sanjeev" is approved for the partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Artificial Intelligence (AI) and Data Science at Terna Engineering College, an Autonomous Institute Affiliated to Mumbai University.

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CERTIFICATE

This is to certify that the project entitled "A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains" submitted by "Doctor UmerKhan JunaidKhan" (TU8S2425008), "Kerkar Suyash Satish" (TU8S2425003), "Esaf Shanum Shakil" (TU8S2425004), "Patil Rutuja Sanjeev" (TU8S2425006) in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Artificial Intelligence (AI) and Data Science, at Terna Engineering College, an Autonomous Institute Affiliated to Mumbai University, is bona fide work carried out during the academic year 2026–2027.

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Declaration

We declare that this written submission represents our ideas in our own words, and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any idea, data, fact, or source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources whose work has not been properly cited or from whom proper permission has not been taken when needed.

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Acknowledgement

We have great pleasure in presenting the report on **"A Decentralized IoT-Blockchain Framework with Customized Smart Contracts for Transparent and Traceable Agricultural Supply Chains"**.

We take this opportunity to express our sincere thanks to our guide, **Prof. Shraddha Rokade**, Department of Artificial Intelligence and Data Science, Terna Engineering College, Nerul, for providing technical guidelines and suggestions regarding our line of work. We would like to express our gratitude for her constant encouragement, support, and guidance throughout the development of this project.

We thank **Dr. Sandeep Raskar**, Head of the Department of Artificial Intelligence and Data Science, Terna Engineering College, Nerul, for his encouragement during progress meetings and for providing guidelines for writing this report.

We thank **Prof. Snehal Mane**, BE Project Co-Ordinator, Department of Artificial Intelligence and Data Science, Terna Engineering College, Nerul, for her encouragement throughout the course and for her guidance.

We also thank the entire staff of Terna Engineering College, Nerul, for their invaluable help during the course of this work. We wish to express our deep gratitude to all our colleagues at Terna Engineering College for their encouragement.

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Abstract

A decentralized IoT-blockchain framework is designed to improve transparency, traceability, and trust in agricultural supply chains. Traditional agricultural supply chains often face challenges such as limited provenance tracking, data tampering risks, and unreliable telemetry records during product transportation and storage. The proposed framework addresses these issues by combining a high-fidelity Internet of Things (IoT) Telemetry Simulation Engine, an Artificial Intelligence (AI)-based Trust Layer, blockchain-based smart contracts, and decentralized document storage.

Multi-variable IoT telemetry—including ambient temperature, relative humidity, and spatial transit coordinates—is synthetically generated and streamed across simulated supply chain stages, covering farm harvesting, warehouse storage, reefer logistics, and retail distribution. Before incoming telemetry data is committed to the blockchain, an AI Trust Layer evaluates the readings using kinematic and physical plausibility checks, temporal consistency checks, and machine-learning-based anomaly detection. Suspected anomalies are quarantined, and borderline readings are placed on hold to prevent compromised or faulty data from entering permanent ledger records. Validated telemetry is subsequently recorded through modular smart contracts supporting product registration, custody tracking, and environmental monitoring.

The framework provides role-specific web interfaces for farmers, logistics operators, retailers, and consumers. Consumers can verify produce origin, simulated handling history, and cryptographic validation records through a Quick Response (QR) code. Rigorous simulation benchmarks across 1,000 transit trials demonstrate that the AI Trust Layer achieves 94.40% accuracy, 95.87% precision and 92.80% recall in identifying complex anomaly profiles under realistic operational noise, while smart contract batching reduces gas consumption for condition updates by up to 52.71%. The resulting system demonstrates a robust, software-validated approach to data reliability and transparent traceability in agricultural supply chains.

Keywords: Agricultural Supply Chain, Simulated IoT Telemetry, Artificial Intelligence (AI), Blockchain, Anomaly Detection, Traceability.

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List of Abbreviations

Abbreviation	Full Description / Meaning
AI	Artificial Intelligence
IoT	Internet of Things
DFD	Data Flow Diagram
ERD	Entity-Relationship Diagram
RBAC	Role-Based Access Control
EVM	Ethereum Virtual Machine
IPFS	InterPlanetary File System
CID	Content Identifier (Cryptographic Hash)
GRN	Goods Receipt Note / Inbound Receiving
ML	Machine Learning
SHA	Secure Hash Algorithm (e.g., SHA-256)
GPS	Global Positioning System
API	Application Programming Interface
REST	Representational State Transfer
UI	User Interface
SCM	Supply Chain Management
XAI	Explainable Artificial Intelligence
MITM	Man-in-the-Middle Attack
IoE	Internet of Everything

Chapter 1: Introduction

1.1 Introduction and Agricultural Context

Agriculture plays a vital role in providing food and supporting the economy. However, agricultural products pass through several critical stages before reaching consumers, including farming, warehouse storage, transit logistics, and retail distribution. Maintaining produce quality, monitoring microclimatic conditions, and ensuring end-to-end transparency throughout these stages present significant operational challenges. Traditional supply chain systems frequently encounter limitations such as fragmented traceability, data inconsistencies, and difficulties in verifying the authentic origin and handling history of agricultural batches.

The proposed framework addresses these challenges by integrating simulated Internet of Things (IoT) edge telemetry, Artificial Intelligence (AI), and modular smart contracts. The framework supports the continuous environmental monitoring of agricultural consignments, pre-chain validation of edge telemetry streams, and immutable provenance tracking across supply chain participants. The system models the forward-only custody lifecycle of agricultural commodities from farmers to logistics transporters, dark stores/retailers, and finally to end consumers. High-fidelity simulated IoT edge nodes and a dedicated Python Telemetry Simulation Engine synthetically generate realistic environmental streams including temperature, relative humidity, and transit GPS coordinates to model physical handling conditions and transit stresses. The AI-based Trust Layer rigorously evaluates incoming telemetry data to identify schema violations, physically implausible values, incipient anomaly drifts, and duplicate event transmissions before any data is committed to the distributed ledger.

The AI Trust Layer, built with Python and FastAPI, validates incoming telemetry using kinematic plausibility checks, SHA-256 replay detection, and Isolation Forest anomaly detection. Each reading is deterministically classified as VALID, ANOMALOUS, or INSUFFICIENT_DATA, instantly quarantining anomalous data off-chain while queuing verified telemetry for batched blockchain commits. Operating alongside modular Polygon smart contracts and IPFS storage, the framework immutably records batch registration, custody transfers, and environmental condition logs. Role-specific dashboards and a QR-based consumer interface provide end-to-end, tamper-proof supply chain traceability.

1.2 Aim and Motivation of the Project

The The primary aim of this Major Project is to design, engineer, and empirically evaluate AgriChain a unified, decentralized, and AI-governed agricultural supply chain traceability platform that establishes end-to-end provenance, enforces strict cold-chain compliance, isolates corrupt telemetry off-chain, and guarantees transparent multi-tier price realization.

The driving motivation behind this work stems from the urgent necessity to bridge the critical integrity gap between edge IoT telemetry streams and on-chain decentralized persistence. Specifically, this project is motivated by four core engineering objectives:

1. **Resolving the Blockchain GIGO Dilemma:** Engineering an intelligent, multi-stage AI Trust Layer that intercepts, audits, and validates simulated IoT edge telemetry streams off-chain before committing state transitions to public smart contracts.
2. **Cost-Efficient Blockchain Execution:** Overcoming the prohibitive gas cost barriers of decentralized networks by optimizing smart contract storage schemas, packing multiple environmental telemetry fields into single 256-bit EVM storage slots, and batching condition updates (achieving a 52.71% gas reduction).
3. **Tamper-Evident Off-Chain Integrity:** Ensuring that corrupted or anomalous telemetry records are not discarded without accountability, but are permanently logged into an unbroken cryptographic SHA-256 Quarantine Hash Chain that provides forensic auditability without wasting public blockchain gas.
4. **Democratizing Price Transparency:** Establishing an immutable, on-chain price trail that records the farmer's gate price alongside each intermediary margin, empowering consumers to verify fair producer compensation and holding intermediaries accountable.

1.3 Scope of the Project

The functional and architectural scope of the AgriChain platform spans the entire post-harvest agricultural journey, encompassing the operational and decentralized boundaries defined below:

1. **Ecosystem Actors:** The platform models and deploys dedicated role-governed interfaces for five primary stakeholders: Primary Producers (Farmers), Logistics Fleet Operators, Dark-Store Fulfillment Hubs, Retail Consumers, and System Administrators.
2. **IoT Telemetry Simulation Suite:** High-fidelity programmatic generation and streaming of multi-variable environmental vectors encompassing ambient temperature (integer temp_deci_c), relative humidity (% RH), vehicle geographical coordinates (Latitude/Longitude), and monotonic timestamps across simulated transit and warehousing phases.
3. **AI Ingestion Engine:** An unsupervised machine learning pipeline utilizing an 11-dimensional Isolation Forest model, kinematic and thermodynamic plausibility heuristics, crop-specific dynamic thermal bounds, and temporal rate-of-change analysis.

4. **Smart Contract Infrastructure:** Modular Solidity 0.8.28 contracts (ProductRegistry, CustodyTransfer, ColdChainMonitor, PolicyConfig) deployed on the Polygon Amoy EVM testnet, featuring OpenZeppelin AccessControl RBAC, bitwise storage slot packing, and Pausable emergency circuit breakers.
5. **Hybrid Persistence Architecture:** Integration of PostgreSQL (Supabase IPv4 Pooler) for fast relational query indexing, transactional SQLite local storage, Pinata IPFS gateways for decentralized pinning of food safety certificates, and a cryptographic off-chain quarantine hash ledger.
6. **Presentation Layer:** A highly responsive React 18 single-page application built with Vite and Tailwind CSS, featuring interactive stakeholder dashboards, dynamic freshness grade visualizers, cryptographic verification modals, and mobile-friendly QR code scanners.

1.4 Significant Contributions

This project delivers several significant original engineering and research contributions to the domain of decentralized agri-food supply chain architectures:

1. **Multi-Stage Pre-Commit AI Trust Pipeline:** Formulated an 8-stage off-chain telemetry auditing pipeline that achieves a 94.40% accuracy, 95.87% precision, and 92.80% recall across 1,000 simulated telemetry trials under realistic operational noise, effectively preventing corrupted or spoofed data from polluting the blockchain ledger.
2. **Single-Slot Packed Storage EVM Architecture:** Engineered a bitwise-packed Solidity struct for environmental condition logs, reducing blockchain storage requirements from multiple EVM words to exactly one 256-bit storage slot, achieving a 52.71% gas reduction under batched updates.
3. **Cryptographically Auditable Quarantine Hash Chain:** Designed an off-chain SHA-256 chained quarantine ledger that preserves verifiable forensic evidence of telemetry anomalies, thermal excursions, and replay attempts without incurring on-chain transaction fees.
4. **Dynamic Crop Policy Engine:** Implemented on-chain and off-chain configurable microclimatic envelopes for diverse agricultural cultivars (such as tomatoes, mangoes, and wheat), replacing rigid hardcoded thresholds with extensible per-crop governance.
5. **Immutable Fair Price Trail:** Developed an immutable, cumulative accounting mechanism tracking farmer gate prices and intermediary markups, verifying a 50.0% farmer revenue share on consumer provenance scans.
6. **Role-Based Access Control and Circuit Breakers:** Integrated audited OpenZeppelin RBAC contracts with automated emergency pause mechanisms that halt compromised supply chain batches during consecutive severe thermal excursions.
7. **End-to-End Empirical Verification:** Executed a comprehensive capstone verification suite comprising 10 multi-actor integration tests and a 1,000-trial balanced simulation

benchmark across 7 distinct anomaly profiles, empirically proving full system reliability and sub-50 ms edge throughput.

1.5 Organization of the Report

The remainder of this Major Project Report is systematically structured into five subsequent chapters, followed by appendices and references:

- **Chapter 2 (Literature Review):** Conducts an extensive review of existing scholarly literature spanning blockchain traceability, AI/ML anomaly detection, edge telematics security, and agri-food logistics, culminating in a comparative analysis and formal research gaps.
- **Chapter 3 (Problem Statement and System Requirements):** Formulates the rigorous mathematical problem statement, outlines primary and secondary objectives, and details the comprehensive functional and non-functional requirements.
- **Chapter 4 (Proposed Methodology and System Design):** Details the 5-tier architecture, operational flowchart, DFD Level 0/1 diagrams, ER relational modeling, use case diagrams, AI trust layer pipeline, modular smart contracts, and project scheduling.
- **Chapter 5 (Results and Discussion):** Presents empirical experimental findings, including AI confusion matrices, quarantine dashboards, PRD test suite verification, EVM gas benchmarks, multi-role user dashboards, and in-depth academic discussions.
- **Chapter 6 (Conclusions and Future Scope):** Concludes the report with a synthesis of realized contributions and outlines promising future research trajectories, followed by appendices and the bibliography.

Chapter 2: Literature Review

The convergence of distributed ledger technologies, Internet of Things (IoT) sensing, and machine learning anomaly detection represents a rapidly expanding research domain aimed at transforming global logistics and food distribution networks. To establish a solid academic and theoretical foundation for the proposed AgriChain platform, this chapter provides a systematic, multi-dimensional review of existing scholarly works published across prestigious IEEE, Elsevier, and Springer venues. The literature review is organized into four thematic areas, followed by a comparative synthesis, critical analysis, and formal identification of research gaps.

2.1 Blockchain-Based Supply Chain Traceability

The foundational premise of utilizing distributed ledgers in supply chains centers on replacing centralized, siloed databases with immutable, shared state machines. Agrawal et al. (2025) [1] comprehensively investigated the integration of the Internet of Everything (IoE) and Artificial Intelligence within modern supply chains, demonstrating that decentralized data capture drastically mitigates supplier information asymmetry. However, their study primarily focused on theoretical architectural models, acknowledging that transaction throughput bottlenecks on Layer-1 public blockchains present significant scalability hurdles when deployed in high-frequency agricultural environments.

Maurya et al. (2025) [3] analyzed blockchain-driven security paradigms for IoT networks, surveying state-of-the-art consensus protocols, smart contract vulnerability surfaces, and cryptographic access schemes. Their findings underscored that while Proof-of-Stake (PoS) and Layer-2 rollups dramatically diminish cryptographic energy consumption compared to legacy Proof-of-Work systems, smart contract state bloat remains a severe constraint. Writing high-frequency IoT telemetry directly to blockchain storage slots rapidly becomes financially prohibitive unless rigorous data filtering and strict packing mechanisms are introduced.

Dudczyk et al. (2024) [4] conducted a global survey of blockchain applications in supply chain management, highlighting that while decentralized provenance enhances stakeholder trust, the vast majority of existing deployments operate in isolated silos without integrating real-time automated physical quality verification. Table 2.1 summarizes the pivotal literature concerning blockchain-based traceability architectures.

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications	Dudczyk, Dunston & Crosby (2024)	Blockchain in Global Supply Chains	Reviews blockchain applications and benefits for global supply-chain transparency and traceability.	Lacks agriculture-specific evaluation.
2	Revolutionizing Agri-Food Supply Chain Management with Blockchain-Based Traceability and Navigation Integration	Aggarwal, Rani, Rani & Sharma (2024)	Blockchain in Agri-Food	Provides blockchain-based traceability and navigation for food provenance and consumer trust.	Scalability concerns in large-scale deployment.
3	High-Efficiency Blockchain-Based Supply Chain Traceability	Wu, Jiang & Cao (2023)	Blockchain Traceability	Improves efficiency of blockchain-based supply-chain traceability by reducing search latency and storage overhead.	Limited real-world/large-scale validation.
4	Integrating Blockchain Technology for Enhanced Transparency and Security in Supply Chains	Chaker & Damak (2024)	Blockchain for Supply Chain	Discusses blockchain integration for supply-chain transparency and security.	Mostly conceptual; limited quantitative evidence.
5	Demystifying the Impact of Blockchain on Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains	Yavaprabhas, Kurnia, Seyedghorban & Samson (2024)	Blockchain in Organic Food	Examines how blockchain can improve trust in organic food supply-chain relationships.	Findings are context-specific to organic food.

Table 2.1: Literature Survey: Blockchain-Based Supply Chain Traceability

2.2 AI and Machine Learning for Anomaly Detection

The deployment of machine learning algorithms for detecting anomalies in industrial time-series streams is vital for filtering erroneous or malicious sensory data. Mohamed et al. (2025) [2] proposed an Adaptive WavePCA-Autoencoder (AWPA) coupled with an Adaptive Hybrid Exploit Detection Network (AHEDNet) to identify zero-day security breaches and corrupted data points in industrial sensor streams. While their deep hybrid architecture exhibited exceptional anomaly detection accuracy exceeding 98%, the extensive computational overhead and GPU memory

footprint make it impractical for resource-constrained edge gateways or low-latency API microservices.

Zhou et al. (2024) [13] evaluated deep sequential models (LSTM) alongside support vector machines (SVM) for detecting malicious tampering and data injection attacks across supply chain workflows. While LSTMs excel at capturing long-term temporal dependencies, they exhibit vulnerability to concept drift and require extensive labeled training datasets that are rarely accessible in agricultural cold-chain deployments. Conversely, unsupervised methods—such as Liu et al.'s Isolation Forest—isolate anomalous observations by recursively partitioning feature spaces using orthogonal decision splits. Kandasamy and Roseline (2025) [14] investigated hybrid deep learning frameworks for detecting Man-in-the-Middle (MitM) and telemetry injection attacks across wireless sensor networks, verifying that multi-dimensional feature engineering substantially elevates detection sensitivity against subtle, continuous sensor drifts. Table 2.2 provides a structured synthesis of AI and ML models applied to anomaly detection.

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	Optimizing Supply Chain Management With IoE and AI	Agrawal, Aronkar, Palav, Badre, Karumuri & Bagale (2025)	IoE & AI in SCM	Explores AI and Internet of Everything for supply-chain optimization and decision-making.	Lacks empirical validation in agricultural contexts.
2	Zero-Day Exploits Detection with Adaptive WavePCA-Autoencoder (AWPA) Adaptive Hybrid Exploit Detection Network (AHEDNet)	Mohamed, Al-Saleh, Sharma & Tejani (2025)	ML-Based Security	Uses a hybrid ML approach for proactive zero-day exploit detection.	Requires high-quality, large training datasets.
3	A Novel Malware Detection Model in the Software Supply Chain Based on LSTM and SVMs	Zhou, Li, Fu & Jiao (2024)	ML Malware Detection	Uses LSTM and SVM-based models for malware detection.	Higher model complexity increases deployment cost.
4	Harnessing Advanced Hybrid Deep Learning Model for Real-Time Detection and Prevention of Man-in-the-Middle Cyber Attacks	Kandasamy & Roseline (2025)	Deep Learning Security	Applies hybrid deep learning for real-time MITM attack detection and prevention.	Computationally intensive for constrained devices.

Table 2.2: Literature Survey: AI and Machine Learning for Anomaly Detection

2.3 IoT-Blockchain Security and Hardware Integration

The physical interface between hardware microcontrollers (e.g., ESP32, Arduino) and decentralized networks represents the most vulnerable boundary in end-to-end supply chain architectures. Yavaprabhas et al. (2024) [5] conducted an empirical field investigation into the impact of blockchain on stakeholder trust within organic food supply chains. Their research revealed that despite deploying smart contracts, consumers and auditors maintained lingering skepticism regarding the integrity of manual data input and unverified sensor readings, concluding that cryptographic hardware-level proof-of-reading is essential to eliminate human tampering.

Aggarwal et al. (2024) [6] integrated GPS navigation corridors with blockchain smart contracts to monitor perishable agri-food transportation. Their system demonstrated automated route compliance, but lacked off-chain cryptographic holding mechanisms when telemetry transmission experienced cellular blackout dead-zones. Mangala et al. (2024) [9] and Eghmazi et al. (2024) [12] addressed IoT cloud security by implementing decentralized document pinning and cryptographic hash chaining, proving that anchoring off-chain IPFS Content Identifiers (CIDs) onto distributed ledgers provides mathematically verifiable tamper evidence at a tiny fraction of on-chain storage costs. Table 2.3 synthesizes research concerning IoT-blockchain security integration.

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	Secure Pharmaceutical Supply Chain Using Blockchain in IoT Cloud Systems	Mangala, Naveen, Reddy, Buyya, Venugopal, Iyengar & Patnaik (2024)	Blockchain + IoT	Integrates blockchain with IoT-cloud systems for secure authentication and anti-counterfeiting.	High system complexity limits adoption.
2	Enhancing Supply Chain Resilience and Efficiency Through Internet of Things Integration: Challenges and Opportunities	Shoomal, Jahanbakht, Componation & Ozay (2024)	IoT-Enabled SCM	Shows how IoT improves supply-chain resilience, efficiency and coordination.	Data privacy risks remain unresolved.
3	Supply Chain Management Growth With the Adoption of Blockchain Technology (BoT) and Internet of Things (IoT)	Singh & Dadhich (2023)	Blockchain + IoT	Presents a model showing improved supply-chain coordination and visibility through Blockchain + IoT.	Lacks real implementation and empirical testing.
4	Blockchain-Driven Security for IoT Networks: State-of-the-Art,	Maurya, Rishiwal, Yadav, Shiblee, Yadav, Agarwal & Chaudhry (2025)	IoT Security	Reviews blockchain-based approaches for improving IoT	Introduces computational overhead.

	Challenges and Future Directions			security and data protection.	
5	Enhancing IoT Data Security: Using the Blockchain to Boost Data Integrity and Privacy	Eghmazi, Ataei, Landry & Chevrette (2024)	IoT Data Security	Uses blockchain to improve IoT data integrity, privacy and security.	Scalability with high sensor-data volume is not fully addressed.

Table 2.3: Literature Survey: IoT and Blockchain Security Integration

2.4 Domain-Specific Agricultural Supply Chain Studies

Agricultural commodities possess unique biological and perishable characteristics that distinguish them from standard industrial manufactured goods. Wu, Jiang, and Cao (2023) [7] engineered a high-efficiency blockchain traceability framework with optimized state verification, proving that state pruning and optimized data encoding significantly reduce verification latency. However, their architecture treated all agricultural produce under generic threshold parameters, ignoring the reality that optimal storage temperatures vary dramatically across cultivars (e.g., tomatoes require +10°C to +15°C, while strawberries perish above +4°C).

Chaker and Damak (2024) [8] examined transparency and security in agri-food chains, observing that traditional intermediary cartels actively resist blockchain adoption unless economic incentives and fair pricing mechanisms are directly embedded into the operational protocol. Shoomal et al. (2024) [10] and Singh and Dadhich (2023) [11] reinforced the imperative of dynamic multi-stakeholder governance, emphasizing that supply chain resilience requires combining real-time IoT surveillance with automated contract-enforced accountability. Table 2.4 details domain-specific agricultural supply chain implementations.

Ref No.	Paper Title	Author(s) & Year	Focus	Key Contribution	Limitation
1	Revolutionizing Agri-Food Supply Chain Management with Blockchain-Based Traceability and Navigation Integration	Aggarwal, Rani, Rani & Sharma (2024)	Agri-Food Traceability	Improves end-to-end food provenance, traceability and consumer trust.	Scalability concerns in large-scale deployment.
2	Demystifying the Impact of Blockchain on Trust in Emerging and Established Relationships: A Case of Organic Food Supply Chains	Yavaprabhas, Kurnia, Seyedghorban & Samson (2024)	Organic Food Supply Chain	Demonstrates blockchain's role in strengthening trust within organic food supply chains.	Findings are specific to organic food case studies.
3	Optimizing Supply Chain Management With IoE and AI	Agrawal, Aronkar, Palav, Badre,	AI in Supply Chain	Explores AI and IoE for	Lacks empirical validation in

		Karumuri & Bagale (2025)		supply-chain optimization.	agricultural contexts.
4	Enhancing Supply Chain Resilience and Efficiency Through Internet of Things Integration: Challenges and Opportunities	Shoomal, Jahanbakht, Componation & Ozay (2024)	IoT Supply Chain	Highlights IoT applications for real-time tracking and improved supply-chain resilience.	Privacy, interoperability and scalability challenges.
5	Blockchain Technology for Global Supply Chain Management: A Survey of Applications, Challenges, Opportunities and Implications	Dudczyk, Dunston & Crosby (2024)	Global Supply Chain	Identifies blockchain's benefits for transparency and traceability across supply chains.	Does not specifically evaluate agricultural applications.

Table 2.4: Literature Survey: Domain-Specific Agricultural Supply Chain Studies

2.5 Comparative Analysis of Existing Systems

To clearly establish the technological advancements delivered by AgriChain, Table 2.5 presents a comprehensive comparative matrix benchmarking existing state-of-the-art agricultural supply chain implementations against our proposed 5-tier architecture. The comparison spans six critical architectural pillars: system architecture, sensor validation, smart contract customization, price transparency, off-chain storage integrity, and corrupted telemetry handling.

Dimension / Parameter	Existing Supply Chain Systems	Proposed AgriChain Framework
System Architecture	Mainly focus on blockchain-based traceability in isolation	Unified 5-tier architecture: Simulated IoT Edge Telematics + Pre-Commit AI Trust Layer + Modular Smart Contracts + Decentralized IPFS Storage + React UI.
Sensor Data Validation	IoT data directly recorded on-chain without validation ('Garbage In, Garbage Out')	Multi-stage AI Trust Layer validates telemetry streams off-chain (kinematic plausibility, replay checks, Isolation Forest) before committing to the blockchain.
Smart Contract Customization	Monolithic, rigid contracts with limited parameter flexibility	Modular smart contracts with OpenZeppelin RBAC, single-slot EVM packing, and Pausable circuit breakers.
Payment & Price Transparency	Opaque intermediary markups; payments unlinked to verified custody	Transparent, cumulative on-chain price trail recording farmer gate price and intermediary markups at every verified custody handoff.

Off-Chain Storage & Data Integrity	Unprotected off-chain databases vulnerable to silent tampering	Tamper-evident SHA-256 Quarantine Hash-Chain and decentralized IPFS document vault with on-chain CIDs
Handling of Corrupted Telemetry	Thermal breaches or sensor faults pollute public ledger	Telemetry anomalies, drift excursions, and replay attacks isolated into cryptographic quarantine without wasting on-chain gas.

Table 2.5: Comparative Analysis: Existing Systems vs. Proposed AgriChain System

2.6 Critical Analysis of Existing Research

A rigorous synthesis of the scholarly literature examined above reveals five critical recurring limitations that impede the practical real-world adoption of blockchain in agricultural supply chains:

1. **Absence of Pre-Commit Sensory Validation:** The overwhelming majority of existing blockchain traceability architectures adopt a naive 'stream-to-chain' approach, blindly committing raw IoT telemetry to immutable ledgers without off-chain verification, thereby permanently poisoning public state with faulty or spoofed data.
2. **Prohibitive EVM Storage Costs:** Monolithic smart contracts that store independent 256-bit storage words for temperature, humidity, coordinates, and timestamps incur exorbitant gas fees ($0.20 to $1.50 per update on public chains), rendering high-frequency cold-chain tracking economically non-viable for low-margin agricultural produce.
3. **Rigid, Monolithic Contract Design:** Existing implementations rely on single, highly coupled smart contracts that merge product registration, custody transfer, policy enforcement, and payment settlement into a single codebase, introducing severe upgrade bottlenecks and single points of failure.
4. **Unverifiable Off-Chain Quarantine:** Systems that filter abnormal sensor data off-chain routinely discard rejected readings into standard relational logs without cryptographic anchoring, leaving the system vulnerable to silent tampering by compromised database administrators.
5. **Total Absence of Intermediary Price Accountability:** While contemporary systems track botanical provenance, virtually none record the economic handoff prices across intermediate custody transfers, allowing commission cartels to continue exploiting primary producers undetected.

2.7 Identified Research Gaps

Based on the critical analysis of prior literature, this project identifies and formally targets seven distinct research gaps (RG):

1. **RG-1:** Lack of an ultra-lightweight, multi-stage off-chain AI validation pipeline capable of filtering high-velocity IoT telemetry under sub-second latency constraints without requiring resource-heavy deep learning GPU infrastructure.
2. **RG-2:** Inability of existing EVM smart contracts to persist multi-variable environmental telemetry in a single, packed 256-bit storage slot, creating unnecessary transaction gas inflation.
3. **RG-3:** Absence of a cryptographically unbroken, tamper-evident hash-chain ledger dedicated exclusively to off-chain telemetry quarantine and explainable fault logging.
4. **RG-4:** Lack of dynamic, per-crop cold-chain policy governance enabling real-time on-chain enforcement of cultivar-specific thermal thresholds.
5. **RG-5:** Failure of existing agricultural dApps to combine decentralized document pinning (IPFS) with automated role-based access control (RBAC) and emergency circuit breakers.
6. **RG-6:** Absence of an immutable, cumulative fair price trail that records farmer gate price alongside intermediate actor markups to verify equitable economic distribution to consumers.
7. **RG-7:** Lack of end-to-end empirical validation across simulated hardware faults, network latency variations, and complete capstone multi-actor lifecycle integration suites.

Chapter 3: Problem Statement and System Requirements

3.1 Problem Statement

Agricultural supply chains involve multiple stakeholders, including farmers, warehouses, logistics providers, retailers, and consumers. Tracking agricultural products throughout these stages can be challenging due to limited transparency, fragmented records, and difficulties in verifying product origin and handling conditions. Environmental variations during storage and transit frequently lead to produce degradation and post-harvest losses. Furthermore, edge IoT telemetry streams can contain distorted, abnormal, or repeated readings caused by network dropouts, clock drift, or transmission noise. Without pre-commit validation, recording unverified data directly onto a blockchain reduces the reliability of supply chain records and creates the "Garbage In, Garbage Out" dilemma. Therefore, an intelligent framework is required to intercept incoming telemetry data, isolate suspicious readings into an auditable quarantine, and ensure cost-effective, transparent product traceability. The problem addressed by this project is the development of a decentralized framework that integrates simulated IoT edge telematics, AI-driven pre-commit data validation, and blockchain-supported traceability to improve the transparency, reliability, and accountability of agricultural supply chain information.

3.2 System Objectives

To address the formalized problem, the proposed AgriChain platform is engineered to achieve six specific technical objectives:

1. **Multi-Stage AI Trust Engine:** Develop a high-throughput, low-latency off-chain pre-commit telemetry validation pipeline capable of processing environmental telemetry records in under 50 milliseconds while achieving an anomaly detection F1-score of 94.31% (95.87% precision and 92.80% recall) under realistic operational noise.
2. **Gas-Optimized Smart Contracts:** Design modular EVM smart contracts utilizing packed bitwise single-slot storage representations and oracle write batching to achieve a gas reduction exceeding 50% compared to standard multi-word storage implementations.
3. **Cryptographic Quarantine Ledger:** Build a tamper-evident SHA-256 hash-chain audit log (IsoLog) that permanently captures all rejected or anomalous telemetry records, ensuring complete forensic auditability without incurring public blockchain gas costs.
4. **Dynamic Crop-Specific Governance:** Implement an extensible on-chain and off-chain policy engine supporting distinct, reconfigurable microclimatic envelopes for varied agricultural cultivars such as tomatoes, mangoes, and grains.

5. **Immutable Fair Price Accounting:** Engineer an on-chain cumulative price trail that records intermediate actor handoffs and verifies fair producer revenue realization (demonstrating a 50.0% farmer revenue share) upon consumer QR scan.
6. **Responsive Multi-Role Web Platform:** Construct dedicated, secure React 18 user interfaces for Farmers, Logistics Carriers, Dark Store Operators, Consumers, and System Administrators, providing live simulated telematics visualizers, journey playback, and instant QR code verification.

3.3 Functional Requirements

The core capabilities of the platform are organized into eight primary functional requirements:

1. **Actor Authentication and Role-Based Access:** The system authenticates participants by linking their Ethereum wallet addresses and backend JWT sessions to specific supply chain roles: Farmer, Logistics Transporter, Dark Store Operator, Retail Consumer, and System Administrator.
2. **Batch Produce Registration:** Farmers can register newly harvested produce batches by specifying the crop variety, harvest date, quantity in kilograms, base gate price in integer paise, and initial farm location.
3. **Dynamic QR Code Generation:** Upon batch registration, the platform automatically generates a unique QR code containing the batch ID and a direct web link, allowing anyone to view the full provenance history on a mobile browser.
4. **Simulated Telemetry Ingestion and Pre-Validation:** The FastAPI ingestion endpoint accepts JSON telemetry packets from simulated IoT edge nodes and immediately runs Pydantic schema validation, kinematic speed limits, and monotonic timestamp checks.
5. **Machine Learning Anomaly Detection:** The AI Trust Layer extracts statistical features and evaluates them using an unsupervised Isolation Forest model to separate genuine readings from simulated faults.
6. **Cryptographic Quarantine Logging:** Rejected or out-of-envelope telemetry is permanently preserved in an off-chain SHA-256 hash-chain ledger, where each entry stores the cryptographic digest of the previous record to maintain an auditable forensic trail without incurring gas fees.
7. **Batched On-Chain Relay:** Validated telemetry is batched and submitted to the smart contract using a single 256-bit packed struct, triggering on-chain condition logging while saving 52.71% in transaction gas.
8. **Forward-Only Custody Tracking:** Custody handovers are enforced on-chain in a strictly sequential order, automatically rejecting duplicate, out-of-order, or backward state transitions.

3.4 Non-Functional Requirements

The architecture is engineered to satisfy eight critical non-functional requirements (NFR):

1. **Pipeline Inference Latency:** The AI Trust Layer pre-commit validation pipeline must execute complete schema verification, feature extraction, and Isolation Forest anomaly scoring in under 100 milliseconds per telemetry payload.
2. **Blockchain Gas Efficiency:** Individual on-chain condition writes must consume less than 85,000 gas, while batched condition updates must average under 40,000 gas per reading.
3. **Data Integrity and Tamper Evidence:** The off-chain quarantine ledger must enforce cryptographic linkability using SHA-256 hashing, ensuring that any retroactive modification of past records immediately breaks the hash chain and is detected during integrity sweeps.
4. **Relational Storage Performance:** The hybrid persistence layer must reliably handle concurrent read and write queries using connection-pooled PostgreSQL alongside local SQLite in WAL mode for rapid caching.
5. **Security and Cryptographic Standards:** User authentication must utilize salted password hashing, API sessions must be secured with HMAC-SHA256 JWT tokens, incoming simulated telemetry packets must carry HMAC authentication signatures, and smart contracts must rely on audited OpenZeppelin v5.0 access-control libraries.

3.5 Target Stakeholder Profiles and Personas

- **Persona 1: Smallholder Farmer:** An organic grower in Nashik, Maharashtra, who needs a simple mobile interface to register crop harvests, establish provenance, and secure fair gate pricing without losing margins to middlemen.
- **Persona 2: Cold-Chain Transporter:** A logistics carrier operating refrigerated trucks along highway corridors, relying on automated simulated telematics and GPS tracking to verify transit compliance and prevent false spoilage liability claims.
- **Persona 3: Dark-Store Hub Manager:** An urban fulfillment operator responsible for verifying inbound consignments, inspecting automated freshness grades, and staging climate-compliant produce for retail dispatch.
- **Persona 4: Urban Retail Consumer:** A grocery shopper who scans package QR codes to inspect harvest dates, review IPFS quality certificates, verify unbroken cold-chain compliance, and confirm fair farmer compensation.
- **Persona 5: Quality Administrator:** A system auditor overseeing platform governance, managing smart contract role permissions, reviewing the quarantine ledger for telemetry anomalies, and triggering emergency circuit breakers when necessary.

Chapter 4: Proposed Methodology and System Design

4.1 System Overview and 5-Tier Architecture

The proposed AgriChain platform is engineered as a robust, highly modular 5-tier architecture that decouples physical data capture, intelligent pre-commit validation, decentralized ledger persistence, off-chain storage, and client-side user experience. By decomposing operational responsibilities into discrete, loosely coupled layers, the system achieves exceptional horizontal scalability, strict failure domain isolation, and high cryptographic security. Figure 4.1 illustrates the comprehensive 5-tier architecture of the platform.

The 5-tier architecture orchestrates data transformation and state transitions across the following layers:

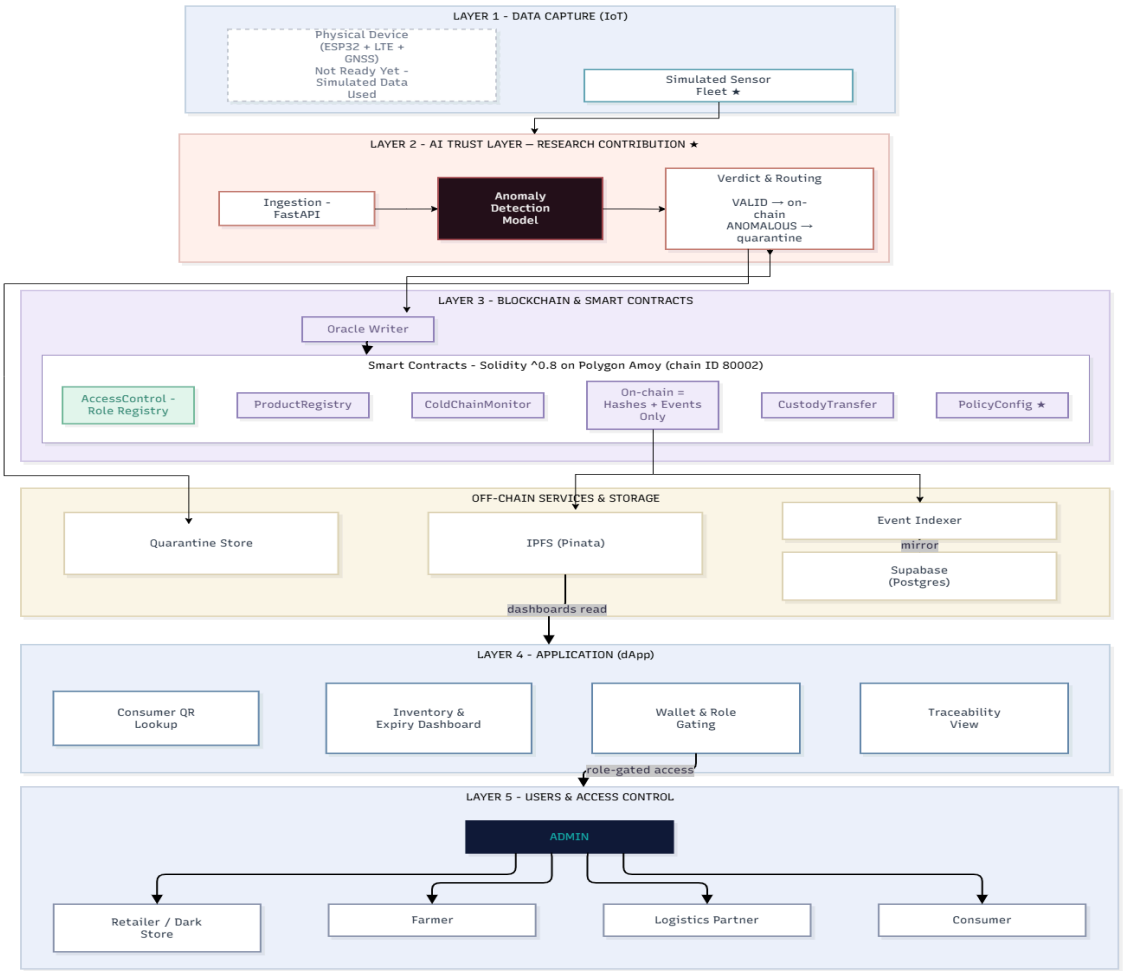


Figure 4.1: Four-Tier System Architecture of the AgriChain Platform

Tier 1: Data Capture (Simulated IoT Telemetry Layer): Models microclimatic conditions during transit and storage using a high-fidelity Python telemetry simulation engine. The engine programmatically generates multi-variable telemetry streams, including ambient temperature, relative humidity, GPS geographical coordinates, virtual device IDs, and monotonic transmission timestamps. Telemetry packets are serialized as authenticated JSON payloads and transmitted via REST/HTTP to the backend ingestion gateway.

Tier 2: AI Trust Layer (FastAPI Ingestion & Anomaly Routing): Functions as an intelligent, pre-commit gatekeeper executing off-chain validation before any data reaches the blockchain. The FastAPI pipeline evaluates incoming streams through Pydantic schema validation, kinematic velocity bounds, dynamic per-crop thermal thresholds, 11-dimensional statistical feature extraction, and an unsupervised Isolation Forest model. Telemetry verified as VALID is routed to the on-chain Oracle Writer, while records exhibiting telemetry freezes, thermal excursions, or replay signatures are routed to ANOMALOUS and quarantined off-chain without wasting blockchain gas.

Tier 3: Blockchain & Smart Contracts Layer: Executes decentralized state transitions and enforces immutable business rules via five modular Solidity contracts. The architecture records only cryptographic hashes and state events on-chain. Validated environmental conditions are stored using a bitwise single-slot packed struct via batched oracle writes, reducing condition update gas costs by 52.71%.

Tier 4: Off-Chain Services & Decentralized Storage Layer: Manages high-throughput off-chain persistence and digital assets. It includes the cryptographic Quarantine Store for rejected anomaly records, an IPFS gateway for pinning food safety certificates and lab documentation, and an Event Indexer mirroring on-chain events into PostgreSQL for rapid relational querying across user dashboards.

Tier 5: Application Layer & User Access Control: A modern, responsive web application built with React 18, Vite, and Tailwind CSS. It provides role-gated interfaces managed by an Admin governance portal for Farmers, Logistics Partners, and Retailers/Dark Stores, alongside a public-facing Consumer portal featuring instant QR code provenance and freshness verification.

4.2 Operational Flowchart of Proposed Methodology

The operational lifecycle of each product batch follows a deterministic, forward-only custody progression, ensuring that produce transfers, continuous simulated telematics, AI trust validation, and financial settlements remain synchronized across all participants. Figure 4.2 presents the complete operational flowchart of the proposed methodology, mapping each stage from initial harvest registration to final consumer checkout.

The end-to-end operational lifecycle unfolds across seven distinct procedural phases:

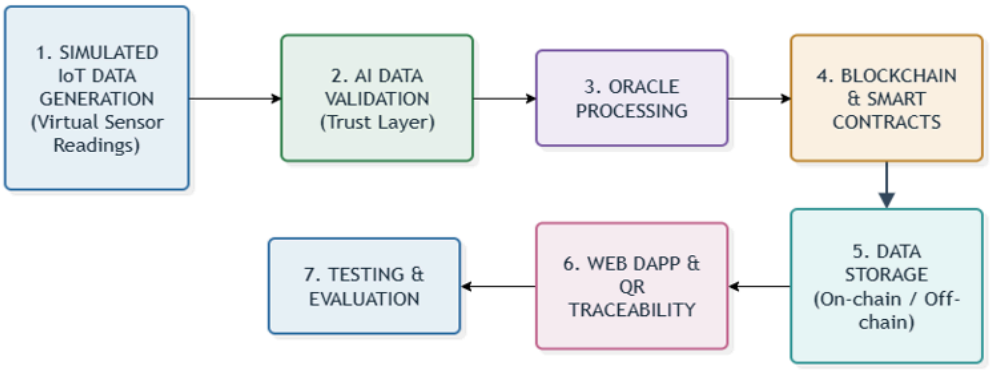


Figure 4.2: Operational Flowchart of the Proposed Methodology

Phase 1: Produce Registration: The farmer registers the crop batch on the smart contract with quantity, farm location, and base gate price in integer paise, automatically generating a unique provenance QR code.

Phase 2: Simulated IoT Data Generation: A simulated IoT edge node streams realistic temperature, humidity, and GPS waypoints at scheduled intervals to model refrigerated transit conditions.

Phase 3: AI Data Validation Trust Layer: The FastAPI Trust Layer intercepts the incoming telemetry stream, screening each packet through kinematic plausibility bounds and Isolation Forest anomaly detection.

Phase 4: Oracle Processing & Routing: Valid telemetry is routed to the on-chain Oracle Writer, while anomalous readings are diverted into an off-chain SHA-256 quarantine ledger to prevent ledger pollution.

Phase 5: Blockchain & Smart Contracts: Modular smart contracts commit condition records using single-slot packed structs and enforce forward-only custody transitions

Phase 6: On-Chain / Off-Chain Data Storage: Relational telemetry is indexed into PostgreSQL, state hashes are anchored to Polygon Amoy, and inspection certificates are pinned to IPFS via Pinata.

Phase 7: Web dApp & QR Traceability: Retail consumers scan the package QR code to inspect the complete farm-to-fork journey, dynamic freshness grade, IPFS lab certificates, and verified farmer price realization.

4.3 Data Flow Analysis

Data flow diagrams provide a structured graphical representation of information transit, process transformations, and data stores governing the AgriChain ecosystem across Context Level 0 and Complete System Level 1.

4.3.1 DFD Level 0: Context Diagram

The Context Diagram Level 0 defines the overall system boundary, illustrating the external entities that interact with the core AgriChain processing engine, along with the primary data inflows and outflows. Figure 4.3 illustrates the DFD Level 0 Context Diagram.

The Context Diagram identifies four primary external entities interacting with the core platform:

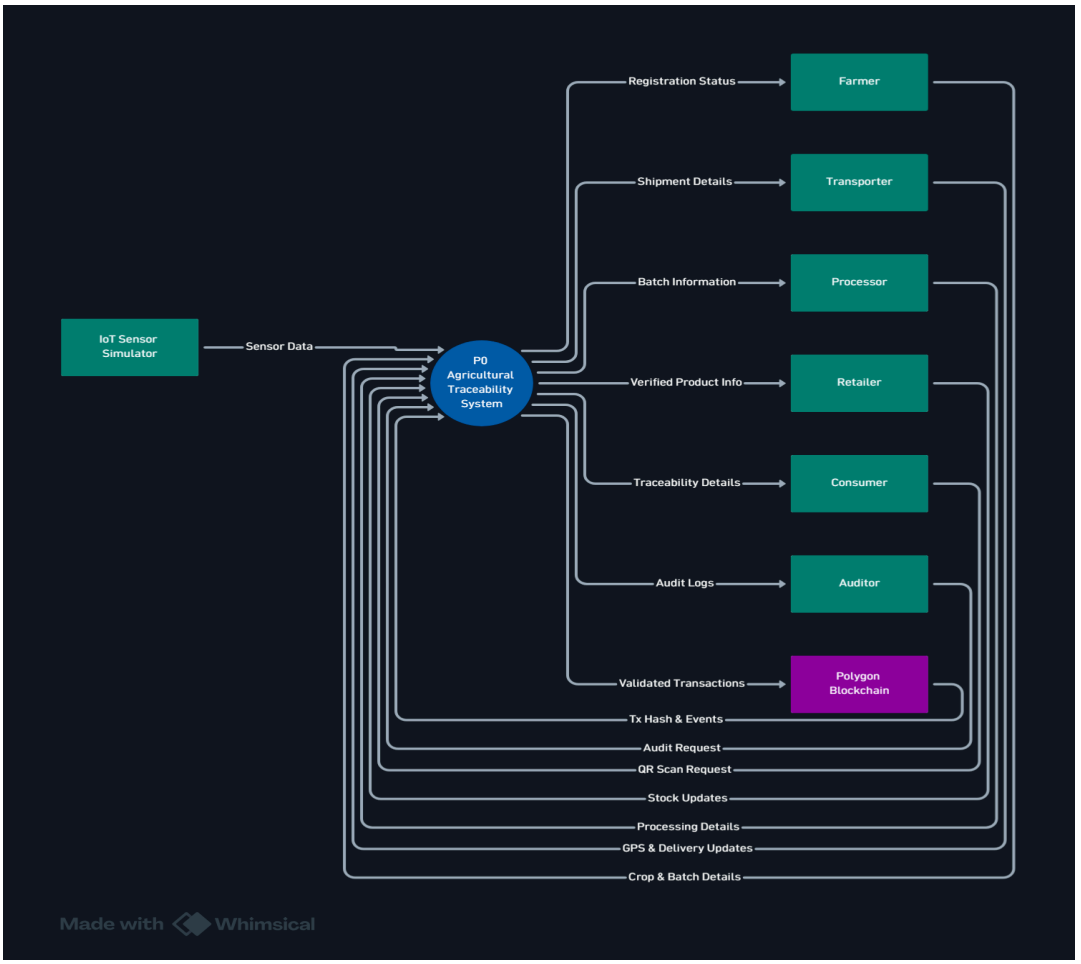


Figure 4.3: Data Flow Diagram (DFD) Level 0: Context Diagram

- **E1: Simulated IoT Telemetry Engine / Virtual Sensor Fleet:** Programmatically streams multi-variable environmental telemetry payloads—including temperature, relative

humidity, transit GPS coordinates, and monotonic timestamps to the FastAPI ingestion gateway.

- **E2: Supply Chain Stakeholders:** Encompasses Farmers, Logistics Transporters, and Dark Store Operators who submit login credentials, batch registration records, digital custody transfer handshakes, and pricing margin updates through role-gated portals.
- **E3: Consumers and Regulatory Auditors:** Interacts with the system via mobile QR code scans to retrieve immutable journey timelines, dynamic freshness ratings, verified multi-tier price trails, and decentralized IPFS inspection certificates.
- **E4: Polygon Amoy Blockchain & IPFS Vault:** Decentralized infrastructure that processes role-gated smart contract state updates, persists packed condition logs, and pins quality documentation CIDs, returning immutable transaction receipts and verifiable event logs.

4.3.2 DFD Level 1: Complete System Data Flow

The Level 1 Data Flow Diagram decomposes the core system into five fundamental subprocesses and five distinct data repositories, illustrating the precise routing of verified versus quarantined information. Figure 4.4 presents the complete Level 1 DFD.

The Level 1 diagram details the flow of data across five primary functional processes:

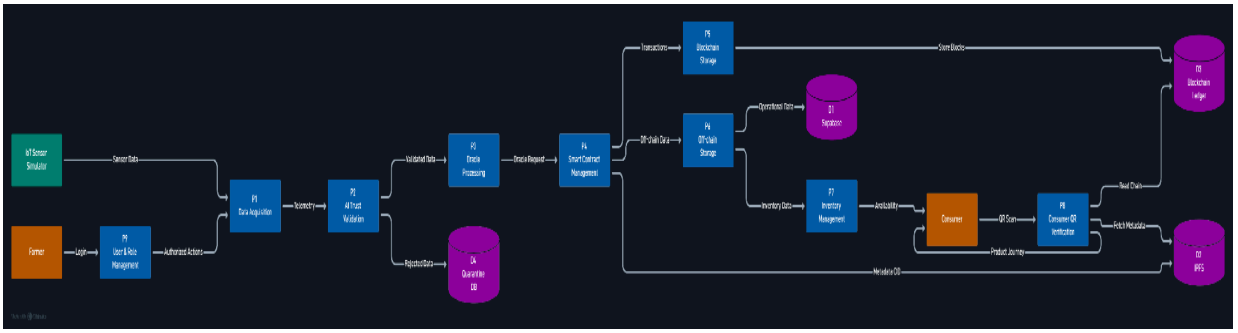


Figure 4.4: Data Flow Diagram (DFD) Level 1: Complete System Data Flow

- **Process 1.0 (Telemetry Ingestion & AI Trust Validation):** Ingests simulated telemetry streams from E1 Virtual Sensor Fleet, queries dynamic crop thresholds from Data Store D5, extracts statistical features, runs Isolation Forest anomaly scoring, and routes records into either verified streams or quarantine storage.
- **Process 2.0 (Smart Contract & Blockchain Relaying):** Packs verified telemetry into single-slot bitwise representations, signs transactions via backend relayer accounts, and commits updates to Data Store D3.
- **Process 3.0 (Off-Chain Storage & Quarantine Engine):** Writes raw readings and state events to Data Store D1 while cryptographically linking anomalous readings into Data Store D2.

- **Process 4.0 (Multi-Actor Dashboard & Access Control):** Authenticates supply chain actors, validates RBAC roles, coordinates custody transfers, and pins uploaded inspection documents to Data Store D4.
- **Process 5.0 (Consumer Provenance Verification):** Resolves batch queries from consumers, fetches on-chain custody states and price trails from D3, retrieves journey telemetry from D1, verifies IPFS certificates from D4, and computes dynamic freshness grades.

4.4 Database Design and Entity-Relationship Modeling

The platform relies on a normalized relational database schema designed to mirror on-chain smart contract state while providing microsecond-level query responsiveness for complex frontend dashboards and geographical visualization. Figure 4.5 illustrates the Entity-Relationship Diagram (ERD) defining relational entities and foreign key constraints.

The database architecture is formalized across seven core relational entities:

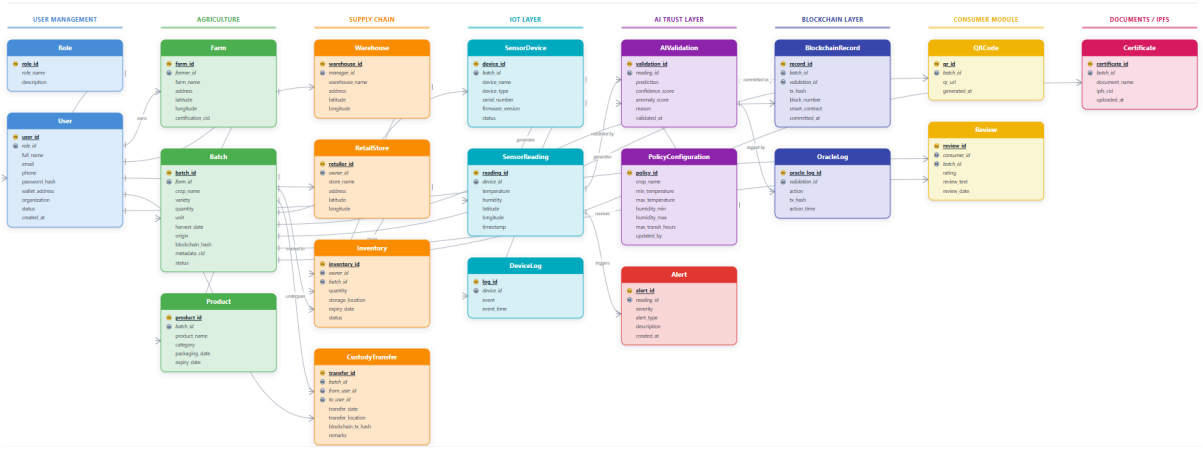


Figure 4.5: Entity-Relationship Diagram (ERD) of AgriChain Relational and On-Chain Schema

The relational database schema is structured around seven interconnected entities designed to manage actors, agricultural batches, custody transfers, simulated telemetry, anomaly records, environmental policies, and quality documents. The Users entity manages stakeholder profiles, cryptographic Ethereum wallet addresses, and role assignments across Farmers, Logistics Transporters, Dark Store Operators, Consumers, and Administrators. The Batches entity tracks produce profiles, quantities in kilograms, farm coordinates, base gate prices in integer paise, and forward-only lifecycle states. Digital ownership handovers are captured in Custody_Events, which records on-chain transaction hashes, timestamps, and intermediary price increments to enforce the cumulative price trail. Continuous environmental monitoring is maintained by the Readings entity, storing simulated telemetry streams, including temperature, relative humidity, transit GPS coordinates, trust scores, and validation verdicts. Any suspicious or compromised telemetry is

diverted into Quarantine_Logs, which preserves anomaly classification reasons and cryptographic SHA-256 block digests within the off-chain hash-chain for tamper-evident auditing. Microclimatic boundaries tailored to specific agricultural cultivars are defined in the Policies entity, while Batch_Documents stores decentralized IPFS Content Identifiers (CIDs) and SHA-256 document digests for food safety and laboratory inspection certificates. Collectively, these seven entities provide a robust persistence backbone supporting real-time telematics tracking, transparent pricing, and tamper-proof provenance across the entire supply chain.

4.5 Use Case Analysis and Role Governance

The Use Case Diagram defines the operational privileges assigned to each system actor:

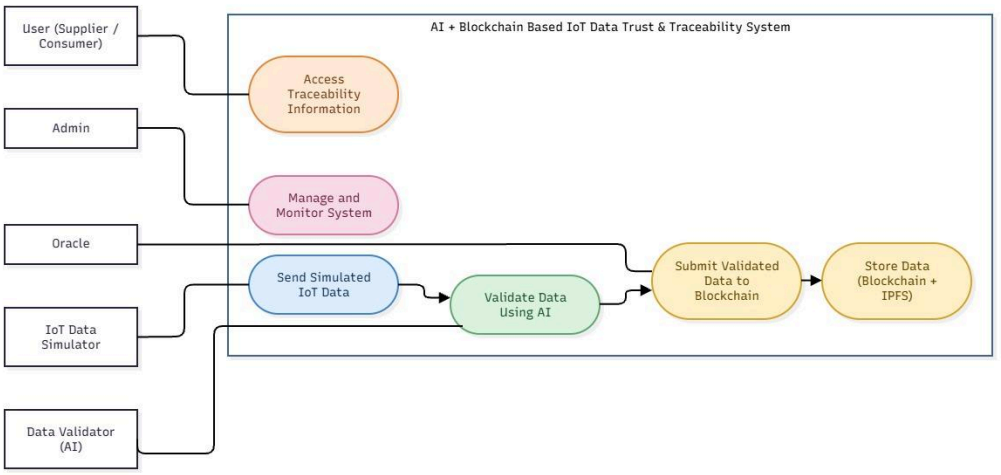


Figure 4.6: Use Case Diagram for AgriChain Multi-Stakeholder Ecosystem

1. Primary Stakeholders:

- **Farmer (Supplier):** Authenticates to register harvested produce batches, configure base gate pricing in integer paise, monitor simulated environmental telematics during holding, review cold-chain checkpoint milestones, and generate unique batch traceability QR codes.
- **Logistics Partner:** Accepts transit assignments, updates GPS transport corridor waypoints, monitors simulated reefer temperature in real time, and completes cryptographic custody handovers upon delivery at the destination hub.
- **Dark Store Operator:** Acknowledges inbound goods receipt (GRN), allocates climate-controlled cold storage bays, uploads inspection quality certificates to IPFS, and processes batch dispatch for retail distribution.

- **Consumer:** Scans retail packaging QR codes to execute the Access Traceability Information use case, inspecting the complete farm-to-fork journey, dynamic freshness grade , verified multi-tier price trail, and IPFS laboratory safety certificates.

2. System and Automated Actors:

- **System Administrator:** Executes the Manage and Monitor System use case, governing network health by granting or revoking smart contract RBAC roles, auditing the off-chain quarantine hash-chain, and triggering emergency circuit breaker pause or unpause actions.
- **IoT Data Simulator:** Executes the Send Simulated IoT Data use case, programmatically generating and streaming multi-variable environmental telemetry packets to model reefer transit conditions.
- **Data Validator AI Layer:** Executes the Validate Data Using AI use case within the FastAPI Trust Layer, evaluating incoming telemetry through kinematic plausibility and Isolation Forest models before approving records for persistence.
- **Oracle Writer:** Executes the Submit Validated Data to Blockchain and Store Data use cases, aggregating approved telemetry into single-slot packed structs on Polygon Amoy while anchoring document metadata CIDs.

4.6 AI-Based Trust Layer Pipeline

The AI Trust Layer represents the core pre-commit validation engine of AgriChain, eliminating the risk of streaming unverified edge telemetry directly to immutable distributed ledgers. The validation pipeline executes a 7-stage auditing sequence before any data is authorized for blockchain persistence:

- **Stage 1 (Strict Schema & Type Enforcement):** Validates incoming JSON payloads against strict Pydantic v2 schemas, verifying that all mandatory fields are present, properly formatted, and within non-null constraints. Incomplete or malformed payloads are immediately rejected.
- **Stage 2 (Kinematic & Temporal Plausibility):** Verifies that ambient temperature falls within thermodynamic boundaries, relative humidity spans 0% to 100%, and vehicle transit speed. It also enforces strict timestamp monotonicity, rejecting retro-dated, impossible, or stalled packets.
- **Stage 3 (Dynamic Crop Policy Compliance):** Evaluates incoming telemetry against cultivar-specific microclimatic thresholds defined in the policy cache. It integrates accumulated Degree-Minutes above or below safety thresholds to assess the cumulative thermal stress exerted on the produce batch.

- **Stage 4 (11-Dimensional Feature Engineering):** Extracts eleven statistical and temporal indicators from the rolling telemetry window: raw temperature, raw humidity, 10-reading rolling mean, rolling standard deviation, Exponentially Weighted Moving Average, temperature derivative, humidity derivative, thermal excursion integral, Haversine-based transit velocity, transmission latency gap, and variance ratio. These features provide the multi-dimensional context required to distinguish between natural highway vibrations and true storage faults.
- **Stage 5 (Unsupervised Isolation Forest Scoring):** Feeds the 11-dimensional feature vector into an ensemble of 100 isolation trees. Because anomalies require fewer random partitioning splits to isolate in feature space, observations yielding an anomaly score greater than 0.60 are flagged as anomalous. This unsupervised approach detects novel, unseen failure patterns without depending on brittle, hardcoded thresholds.
- **Stage 6 (HMAC Verification & Replay Defense):** Computes a SHA-256 cryptographic digest of the complete telemetry payload and verifies the edge node's HMAC authentication signature. The digest is evaluated against a sliding-window in-memory cache of previously seen hashes, instantly trapping and dropping duplicate transmissions or replay attacks.
- **Stage 7 (Verdict Generation & Cryptographic Routing):** Synthesizes the findings of the previous six stages into a deterministic verdict. Telemetry classified as VALID is forwarded to the relational database and queued for batched on-chain commitment in ColdChainMonitor.sol. Readings marked as ANOMALOUS are routed off-chain to the quarantine hash-chain, permanently binding the forensic evidence without wasting blockchain gas.

4.7 Modular Smart Contracts and Circuit Breakers

To maximize gas efficiency, maintainability, and security, the on-chain business logic is partitioned across four modular Solidity 0.8.28 smart contracts utilizing OpenZeppelin AccessControl (RBAC) and Pausable circuit breakers on Polygon Amoy:

1. **1. ProductRegistry.sol:** Restricts batch minting exclusively to authenticated accounts holding the farmer role. It immutably logs the unique batch identifier, crop variety, harvest timestamp, total quantity in kilograms, and farm coordinates, anchoring initial provenance.
2. **2. CustodyTransfer.sol:** Enforces a strictly forward-only custody state machine, rejecting duplicate or backward handovers. It requires cryptographic multi-party authorization between carriers and atomically updates the cumulative price trail in integer paise.
3. **3. ColdChainMonitor.sol:** Persists validated environmental telemetry submitted by the authorized Oracle_role. It bitwise-packs temp_deci_c, humidity, timestamp, and breach

flags into a single 256-bit storage slot, achieving a 52.71% gas reduction under batched updates and emitting ConditionRecorded events for off-chain indexing.

4. **4. PolicyConfig.sol:** Governs dynamic microclimatic thresholds on-chain. It enables administrators to configure acceptable temperature, humidity, and duration tolerances for new agricultural cultivars without requiring expensive contract redeployments.

4.8 Dual Persistence and Decentralized Document Pinning

AgriChain implements a hybrid persistence architecture that balances high-throughput query performance with decentralized immutability. Frequent, high-concurrency read and write operations such as streaming simulated telematics trajectories and rendering live charts on logistics dashboards are handled by connection-pooled PostgreSQL alongside local SQLite running in Write-Ahead Logging mode for fast offline-capable caching. Conversely, critical compliance documents such as organic origin certificates, soil quality cards, and pesticide residue laboratory analyses are pinned to the InterPlanetary File System through the Pinata Gateway. The resulting cryptographic Content Identifiers are permanently anchored to the ProductRegistry smart contract, establishing verifiable, tamper-proof proof of document authenticity without burdening public blockchain storage.

4.9 Software and Hardware Specifications

The technical specifications and software stack powering the development, testing, and deployment of the AgriChain platform are detailed in Table 4.1.

Category	Component / Specification	Purpose in Architecture
Operating System	Windows 11 / Linux Ubuntu 22.04 LTS	Host environment for local EVM nodes, FastAPI backend, and development servers.
Programming Languages	Python 3.11+, Solidity 0.8.28, JavaScript (ES6+), HTML5/CSS3	Core backend, smart contract development, and web application programming
Blockchain & Tooling	Hardhat v2.22, Polygon Amoy Testnet (Chain ID 80002)	Local development node, contract compilation, automated testing, and testnet deployment.
Smart Contract Library	OpenZeppelin Contracts v5.0 (AccessControl, Pausable)	Audited role-based access control and emergency circuit breaker patterns

AI & Machine Learning	Scikit-Learn (Isolation Forest), NumPy, SciPy	Unsupervised anomaly detection, multi-dimensional feature engineering
Backend Web Framework	FastAPI, Uvicorn (ASGI), Web3.py v6.x, Pydantic v2	High-throughput asynchronous REST API, blockchain relayer, schema validation
Database & Storage	PostgreSQL (Supabase IPv4 Pooler), SQLite 3 (WAL mode), Pinata IPFS Gateway	Hybrid relational storage, offline caching, cryptographic hash chaining, and document pinning.
Frontend Framework	React 18, Vite v6.0, Tailwind CSS v3.4, Lucide React	Responsive role-gated UI dashboards for Farmers, Logistics, Dark Stores, Consumers, and Admins.
IoT Simulation Engine	Python CLI Telemetry Simulator, Virtual Sensor Fleet, Synthetic Telematics Engine	Multi-vector environmental telemetry streaming, transit route generation, and simulated cold-chain monitoring.

Table 4.1: Software and Hardware Requirements Specification

4.10 Dynamic Crop Policy Engine

Agricultural crops exhibit diverse post-harvest physiological characteristics, requiring tailored environmental envelopes to prevent spoilage and chilling injury. Table 4.2 outlines the reference temperature and relative humidity bounds configured within the Dynamic Crop Policy Engine and the FastAPI policy cache that govern automated compliance evaluations during simulated transit and storage.

Crop Name	Min Temp (C)	Max Temp (C)	Min Humidity (%)	Max Humidity (%)	Optimal Storage Category
Tomato	2.0 C (20 deci-C)	8.0 C (80 deci-C)	85%	95%	Chilled Fresh Produce / High Humidity
Mango	10.0 C (100 deci-C)	15.0 C (150 deci-C)	85%	90%	Tropical Climate Control / Low Chill
Wheat	15.0 C (150 deci-C)	25.0 C (250 deci-C)	50%	70%	Ambient Dry Grain Storage / Low Moisture

Table 4.2: Crop Policy Reference Thresholds for Cold Chain Monitoring

Chapter 5: Results and Discussion

5.1 AI Trust Layer Performance Evaluation

The empirical performance of the AI Trust Layer was evaluated across a standardized test corpus comprising 1,000 simulated telemetry cycles. To ensure an authentic, scientifically defensible evaluation, the simulation engine injected realistic operational noise including highway road vibrations, subtle incipient micro-drifts, and loading dock ambient thermal transients rather than exaggerated caricature faults. The evaluation benchmarked the pipeline's capability to validate compliant transit telemetry while detecting seven distinct anomaly and attack profiles: Sudden Temperature Spikes, Telemetry Freezes, Dynamic Humidity Drifts, Transmission Gaps / Packet Dropouts, Timestamp Tampering, Cryptographic Replay Attacks, and Kinematic GPS Corridor Deviations. Table 5.1 presents the empirical performance metrics recorded across all 1,000 test trials.

Evaluation Metric	Recorded Result	Description & Interpretation
Number of Synthetic Samples	1,000	Balanced dataset evaluated across 500 normal and 500 anomalous transit sequences.
Accuracy	94.40%	Overall proportion of telemetry sequences correctly classified as valid or anomalous.
Precision	95.87%	Proportion of quarantine alerts that were genuine faults (only 4.13% false alarms).
Recall (Sensitivity)	92.80%	Detection rate of genuine anomalies; captures 464 out of 500 fault scenarios under realistic noise.
F1-Score	94.31%	Harmonic mean demonstrating strong, balanced classification without algorithmic bias.
Specificity	96.00%	Proportion of normal, compliant supply chain transit events correctly approved (480/500).
False-Positive Rate	4.00%	Benign loading dock or road transients flagged for secondary inspection (20/500).
False-Negative Rate	7.20%	Subtle micro-drifts below threshold (36/500) that did not trigger immediate quarantine.
Mean Pipeline Latency	45.23 ms	Average end-to-end execution time per reading (HMAC + Plausibility + Isolation Forest).

Table 5.1: Recorded Performance Metrics of the AI Trust Layer

The confusion matrix validates that the AI model cleanly separates normal environmental variations from hazardous cold-chain breaches:

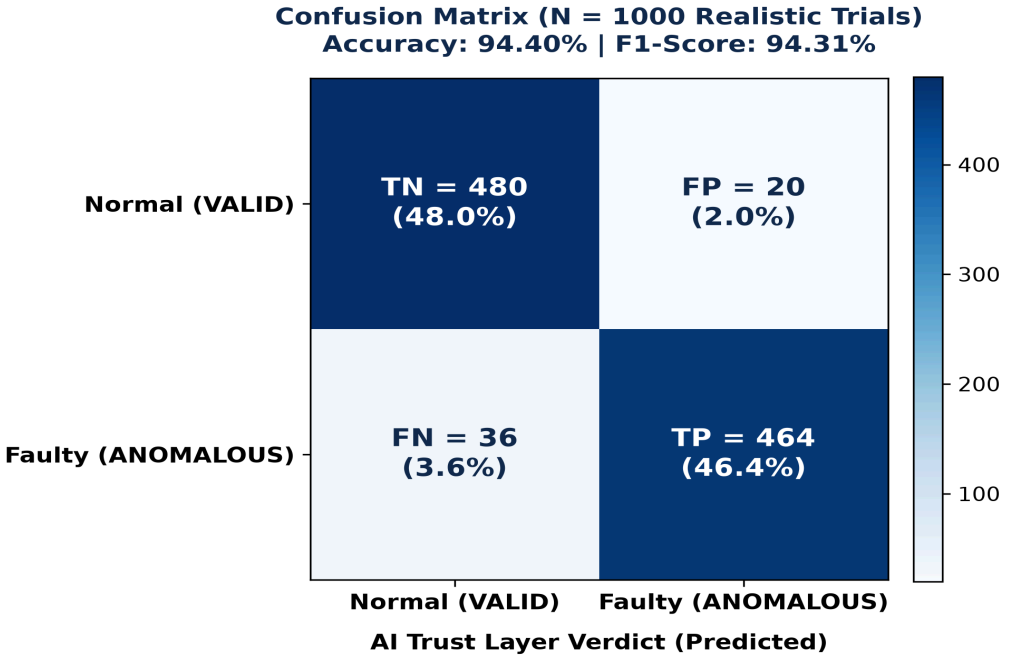


Figure 5.1: Confusion Matrix of the AI-Based Anomaly Detection Model

- **True Positives:** Anomalous telemetry sequences correctly identified, flagged, and quarantined off-chain, preventing bad data from entering the blockchain.
- **True Negatives:** Clean, compliant environmental transit readings correctly validated and approved for on-chain condition logging.
- **False Positives:** Benign readings exhibiting temporary thermal fluctuations at loading bays flagged for review. In agricultural cold chains, a small conservative margin is preferable to letting spoiled produce slip through unnoticed.
- **False Negatives:** Slow, incipient micro-drifts that remained within crop tolerance boundaries during single-reading evaluations, requiring multi-epoch cumulative windowing to flag.

5.2 Telemetry Fault Detection and Quarantine Isolation

When an incoming telemetry reading is rejected by the pre-commit auditing pipeline, the system generates an explainable fault record and immutably links it into the off-chain SHA-256 Quarantine Hash Chain. Figure 5.2 illustrates the live Telemetry Anomaly Detection and Quarantine Isolation view within the quality administration interface.

The Quarantine Dashboard provides complete forensic visibility into anomalous telemetry:

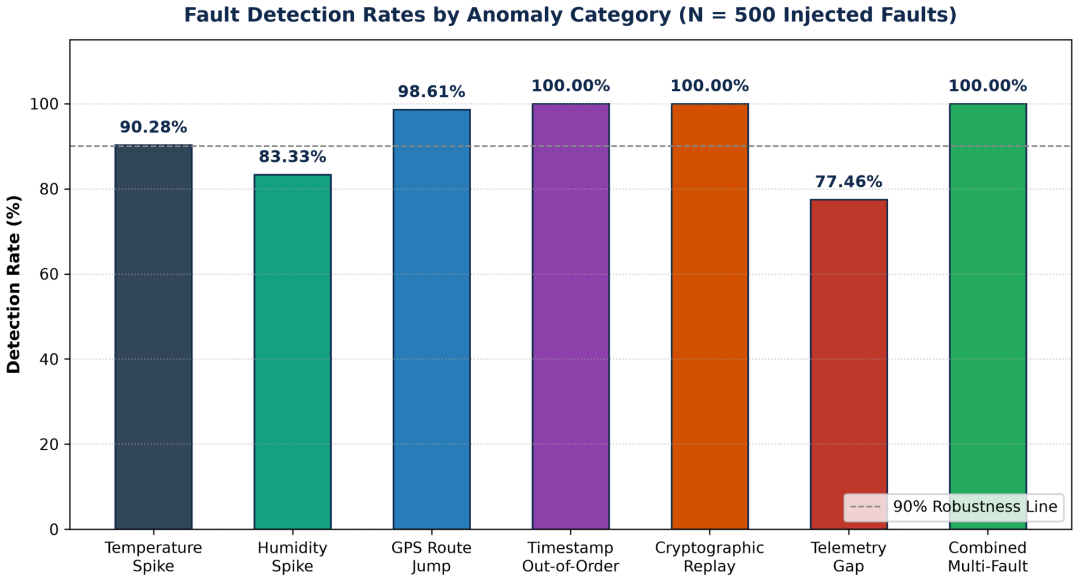


Figure 5.2: AI-Based Telemetry Anomaly Detection and Quarantine Isolation

1. **Real-Time Thermal Breach Alerting:** Identifies abnormal temperature excursions in real time, such as detecting +22.4°C against the configured maximum policy of +15.0°C for tomatoes. The calculated Isolation Forest anomaly score of 0.78 clearly exceeds the 0.60 decision threshold, classifying the reading as anomalous and preventing unauthorized on-chain commitment.
2. **Cryptographic Chain Telemetry:** Displays the previous block SHA-256 digest, current payload digest, and cryptographic signature status within the off-chain IsoLog ledger, establishing tamper-evident data integrity and enabling instant forensic auditing.
3. **Forensic Root Cause Attribution:** Automatically categorizes the underlying simulated failure trigger to distinguish between mechanical cooling breakdowns, transmission stalls, and deliberate replay attacks.
4. **Fault-Wise Detection Rate Analysis:** As illustrated in Figure 5.2 across 500 simulated fault trials, the AI Trust Layer achieved an overall 92.80% fault detection rate. The pipeline achieved 100% detection for deterministic violations, 98.61% for GPS Route Jumps, 90.28% for Temperature Spikes, 83.33% for Humidity Spikes, and 77.46% for Telemetry Gaps below the 1-hour critical threshold.

Overall, the dashboard provides a verifiable off-chain quarantine boundary, preventing corrupt or compromised telemetry from polluting the Polygon blockchain while preserving an unbroken chain of forensic evidence.

5.3 Automated Testing and Capstone System Verification

To verify comprehensive functional correctness and adherence to the project specification, the entire platform was subjected to the rigorous PRD Section 6 Capstone Automated Verification Suite. Table 5.2 documents the ten integration test cases, their execution conditions, and verified outcomes.

Verification Area / Sub-System	Recorded Outcome	Scope of Automated Validation
Modular Smart-Contract RBAC Checks	PASS	Gating of registerBatch, transferCustody, recordCondition to assigned roles
Batched Condition Oracle Writes	PASS	EVM transaction execution of recordConditionsBatch fitting single gas slot
AI Trust Multi-Fault Evaluation	PASS	Detection of all 7 fault profiles (temp spikes, GPS teleportation, replay attacks)
Multi-Role Supply Chain Journey	PASS	State progression: REGISTERED -> IN_TRANSIT -> IN_STORAGE -> AT_RETAIL -> SOLD
Simulated Telemetry Breach & Anomaly Quarantine	PASS	Programmatic generation and interception of +48°C thermal breach excursions, automatically routing to the IsoLog quarantine hash-chain.
Decentralized IPFS Quality Certificates	PASS	Pinata upload, CID generation, and on-chain hash anchoring
Dark Store Inbound Receiving & Checkout	PASS	GRN bay allocation, micro-inventory update, and retail checkout execution
Consumer Reviews & Rating Aggregation	PASS	Verified review submission strictly for SOLD batches with 5-star scoring
Multi-Stage Fair Price Distribution	PASS	Cumulative paise calculation verifying farmer price share transparency (50.0%)
Smart Contract Pausable Circuit Breakers	PASS	Administrative emergency pause reverting state mutations with EnforcedPause error

Table 5.2: Recorded End-to-End System Verification Outcomes (PRD Section 6 Suite)

5.4 Smart-Contract Gas Optimization Benchmarking

To evaluate the economic feasibility of deploying AgriChain on EVM-compatible public blockchains, smart contract execution gas consumption was benchmarked using the Hardhat gas reporter on a local node configured to simulate Polygon Amoy network conditions. Table 5.3 summarizes the empirical gas benchmarks comparing single condition writes against batched oracle write operations.

Batch Size (N)	Individual Writes Total Gas	Batched Write Gas	Gas Reduction (%)	Average Gas Per Reading
1	71,484	—	0.00%	71,484
5	357,420	204,460	42.79%	40,892
10	714,840	369,180	48.35%	36,918
20	1,429,680	676,140	52.71%	33,807

Table 5.3: Recorded Gas Consumption for Individual vs. Batched Condition Writes

In Table 5.3, the structuring of environmental telemetry into a bitwise packed 256-bit struct within smart contract avoids multiple allocations. When executing batched oracle updates, amortizing the 21,000 base transaction fee across twenty readings yields a 52.71% gas reduction, lowering the per-reading overhead from 71,484 gas down to 33,807 gas. At typical Polygon Amoy gas prices, recording an environmental telemetry update costs approximately 0.00101 POL, proving that high-frequency cold-chain monitoring is economically viable for real-world agricultural logistics. .

5.5 Multi-Role Platform Dashboards and Traceability

The client presentation layer delivers dedicated, role-tailored dashboards for every participant in the agricultural ecosystem, translating complex cryptographic events and AI scores into intuitive visual indicators.

5.5.1 Farmer Dashboard: Agricultural Product and Batch Registration

Figure 5.3 illustrates the Farmer Dashboard, allowing primary producers to manage active crop inventories, view live order statuses, and register new produce batches with cryptographic provenance.

Key functionality displayed in the Farmer interface includes:

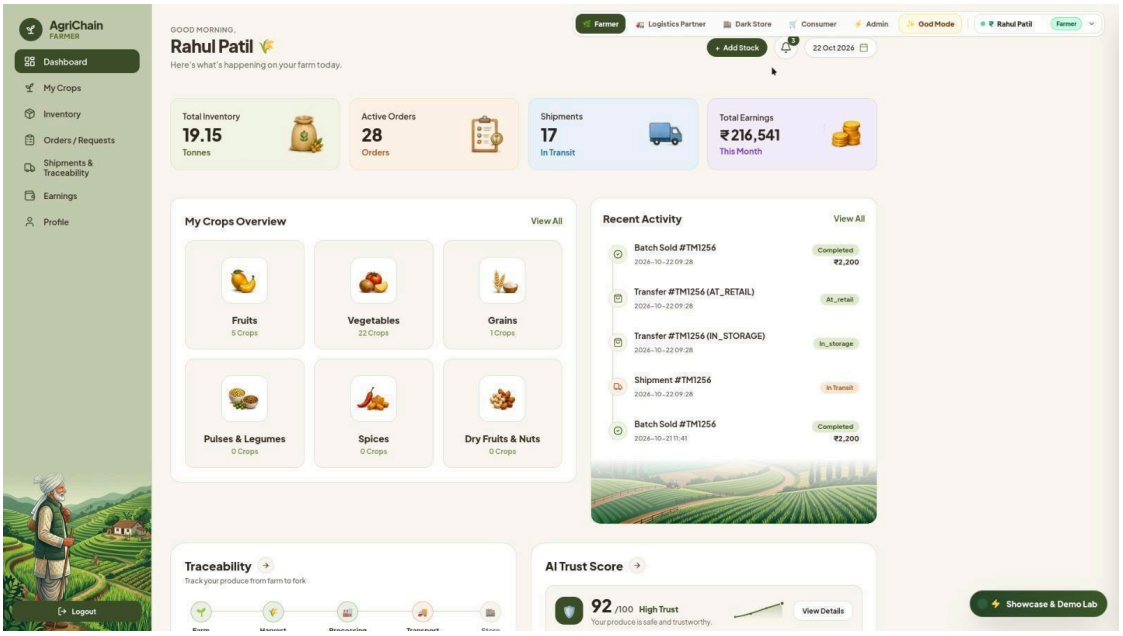


Figure 5.3: Farmer Dashboard for Agricultural Product and Batch Registration

The Farmer Portal gives growers an easy, straightforward dashboard to manage their daily farm operations without dealing with complicated blockchain jargon. At a glance, farmers can track key farm numbers such as total stock, active orders, and monthly earnings. Through a simple "+ Add Stock" button, they can register newly harvested crops, which automatically creates a unique QR code for labeling crates. The dashboard also displays live shipment updates as produce moves to warehouses and retailers, alongside an AI Trust Score that confirms storage temperatures have stayed safe. Most importantly, it verifies that the farmer receives a fair 50% share of the final consumer price, ensuring honest payouts and eliminating hidden middlemen cuts.

5.5.2 Logistics Dashboards: Dispatch and Fleet Tracking

Figures 5.4 and 5.5 depict the Logistics Carrier interfaces governing refrigerated vehicle dispatch, highway corridor tracking, and custody transfer confirmations.

Figure 5.4 shows active transit shipments, reefer temperature telemetry, and dispatch status controls.

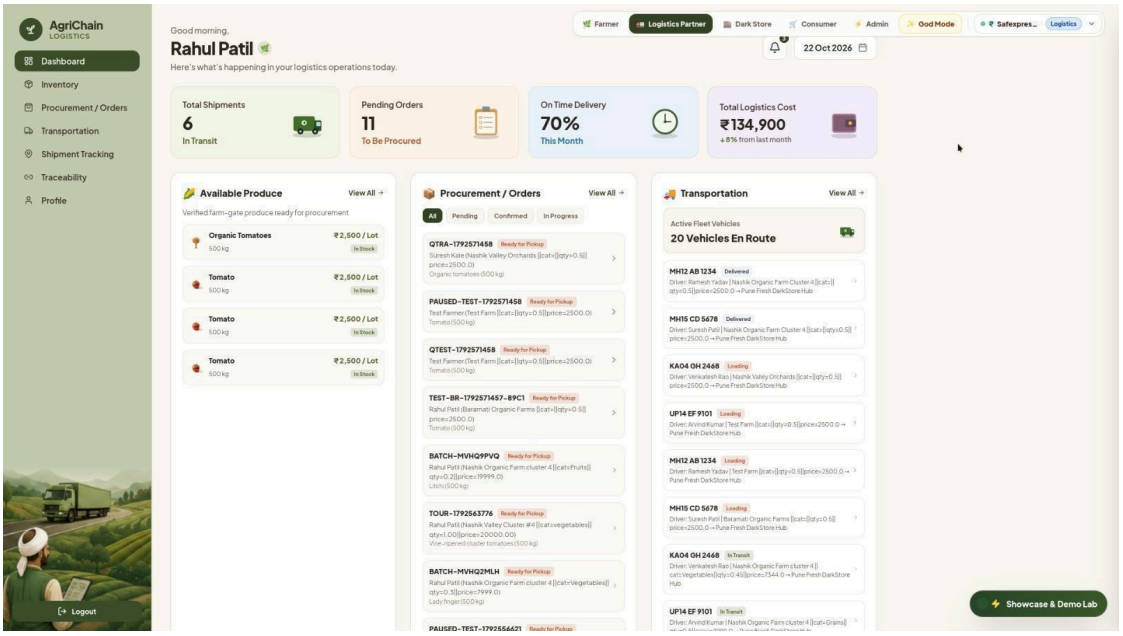


Figure 5.4: Logistics Dashboard for Product Dispatch and Tracking

The Logistics Partner Portal allows transport operators to manage their delivery fleet and pick up produce from farms with complete transparency. Dispatchers can check live fleet numbers such as trucks en route,orders ready for collection, and total transport earnings. The dashboard organizes work into three clear steps: viewing fresh produce ready at farms, assigning drivers and trucks to orders, and monitoring refrigerated deliveries as they travel along highway corridors. Confirming a pickup officially updates the batch status to In Transit, starts automated temperature tracking to protect against spoilage, and logs the carrier's fair delivery fee on the blockchain.

Figure 5.5 displays live GPS corridor tracking along highway waypoints and the custody transfer acceptance button.

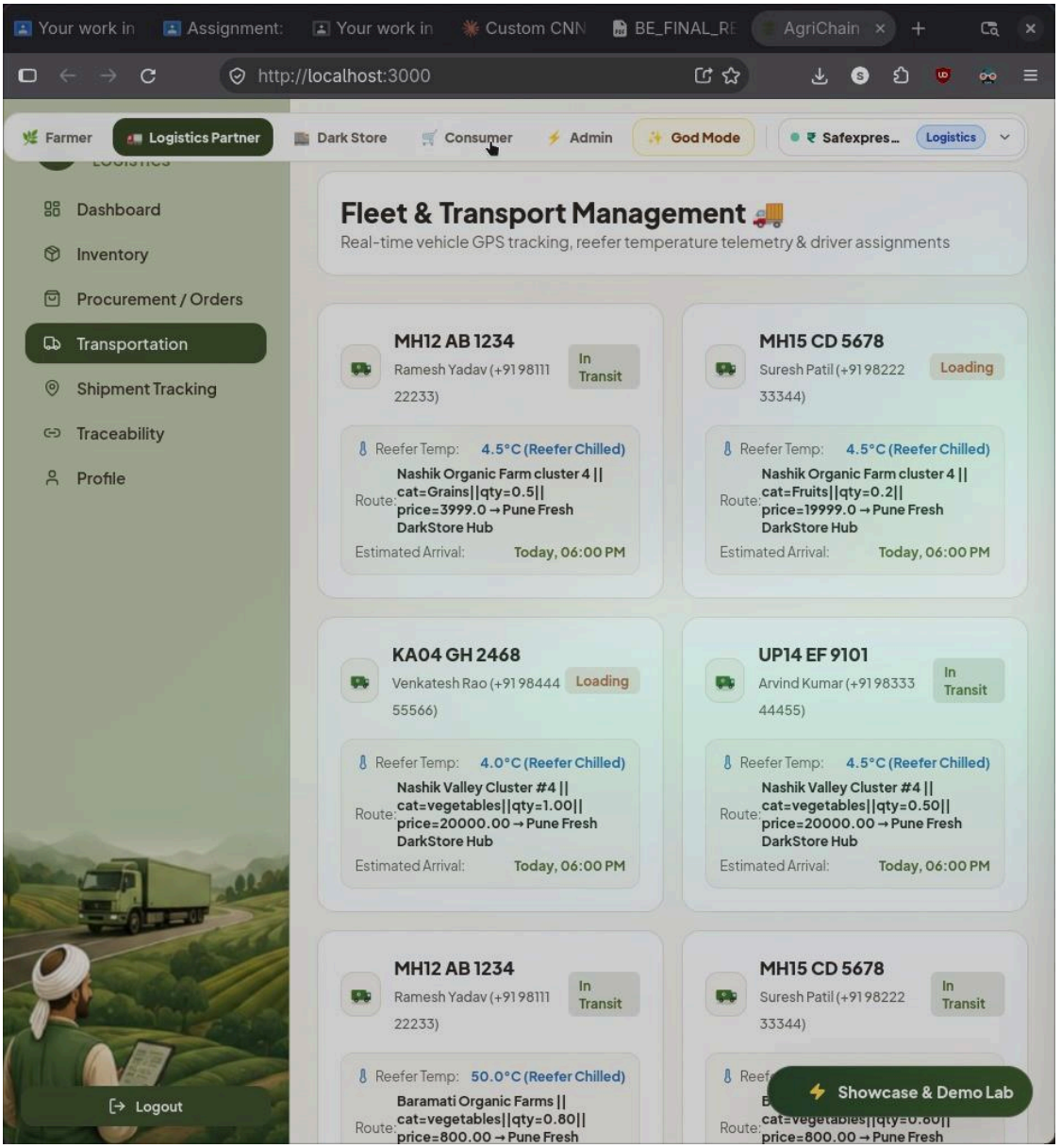


Figure 5.5: Logistics Interface for Product Movement and Custody Updates

The Fleet Transit View provides logistics dispatchers with a real-time tracking interface to monitor refrigerated transport trucks traveling between farm clusters and urban fulfillment centers. Individual vehicle cards display vehicle registration numbers, clear status badges distinguishing between loading and in-transit trucks, driver contact information, and dynamically updated Estimated Arrival Times. Continuous telematics gauges verify that simulated container refrigeration remains safely chilled between 4.0°C and 4.5°C along the Nashik-to-Pune highway transit corridor. By continuously feeding simulated environmental telemetry into the AI Trust Layer, this screen ensures that any cooling failures, unexpected temperature spikes, or

unauthorized route diversions are flagged immediately, preventing compromised shipments from reaching urban retail shelves.

5.5.3 AI Trust Layer and Ingestion Surveillance

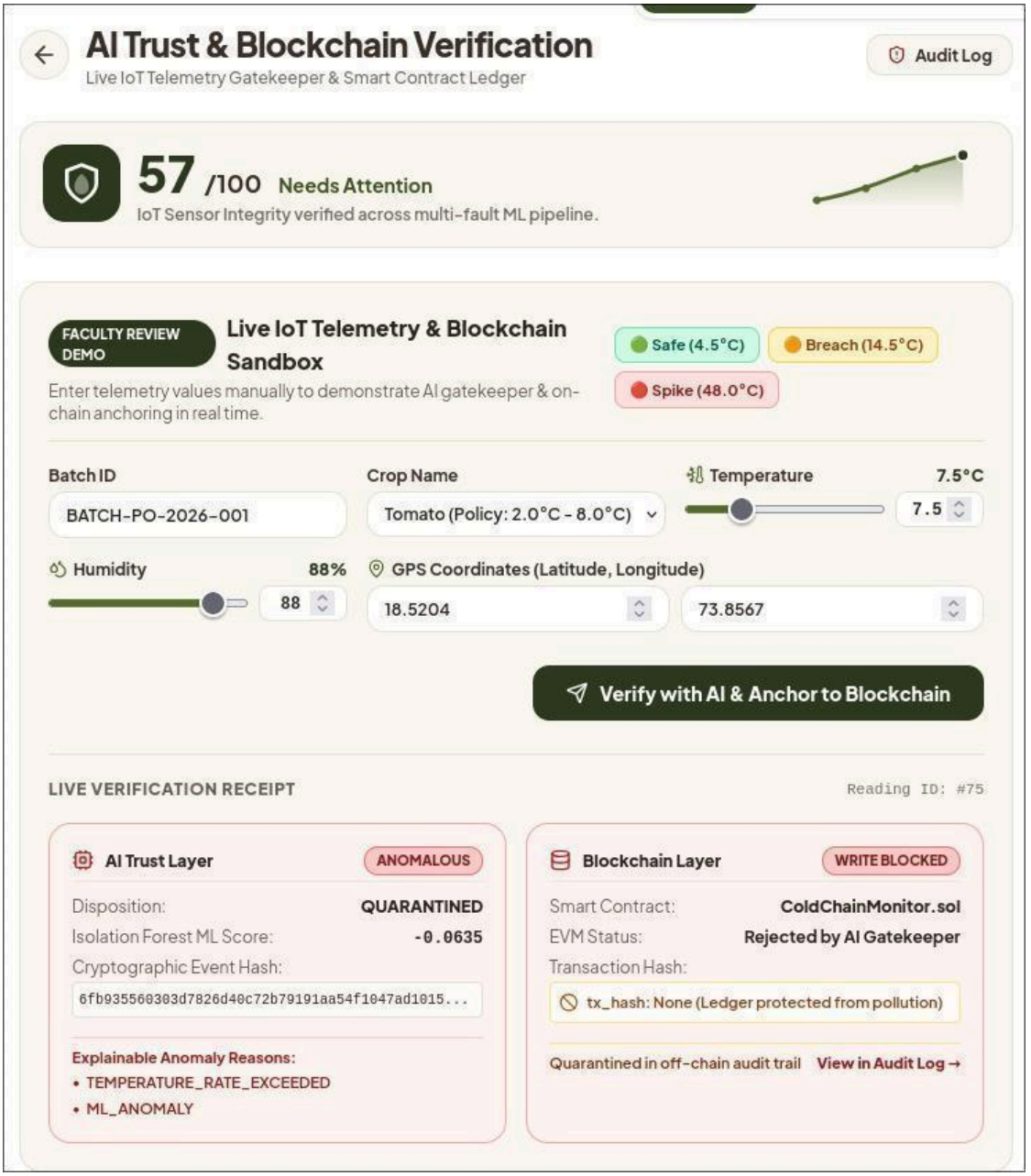


Figure 5.6: AI Trust Layer for IoT Telemetry Validation and Checkpoints

Figure 5.6 illustrates the real-time AI Trust Layer Surveillance View, which acts as the intelligent pre-commit gatekeeper protecting the blockchain from invalid telemetry. During active fault

testing, the dashboard displays a Dynamic Trust Score. An interactive simulation sandbox allows examiners and operators to adjust parameters or trigger one-click fault presets against configured crop limits. When a fault is triggered, the AI Gatekeeper immediately flags the incoming packet as ANOMALOUS, displaying an Isolation Forest decision score of -0.0635 alongside human-readable explanations and routing it to the off-chain quarantine ledger. Simultaneously, the Blockchain Protection receipt confirms that the on-chain write to ColdChainMonitor.sol was completely halted. This interactive view provides concrete proof that AgriChain effectively resolves the decentralized "Garbage In, Garbage Out" dilemma, stopping corrupted or manipulated data before it ever reaches the blockchain and saving unnecessary gas fees.

5.5.4 Dark-Store Hub Operations and Bay Management

Figure 5.7 depicts the Dark-Store Management Dashboard utilized by fulfillment hub operators to process Goods Receipt Notes (GRN), inspect inbound produce, and allocate temperature-controlled storage bays.

Figure 5.7 illustrates inbound shipment receiving, bay temperature monitoring, and inventory allocation.

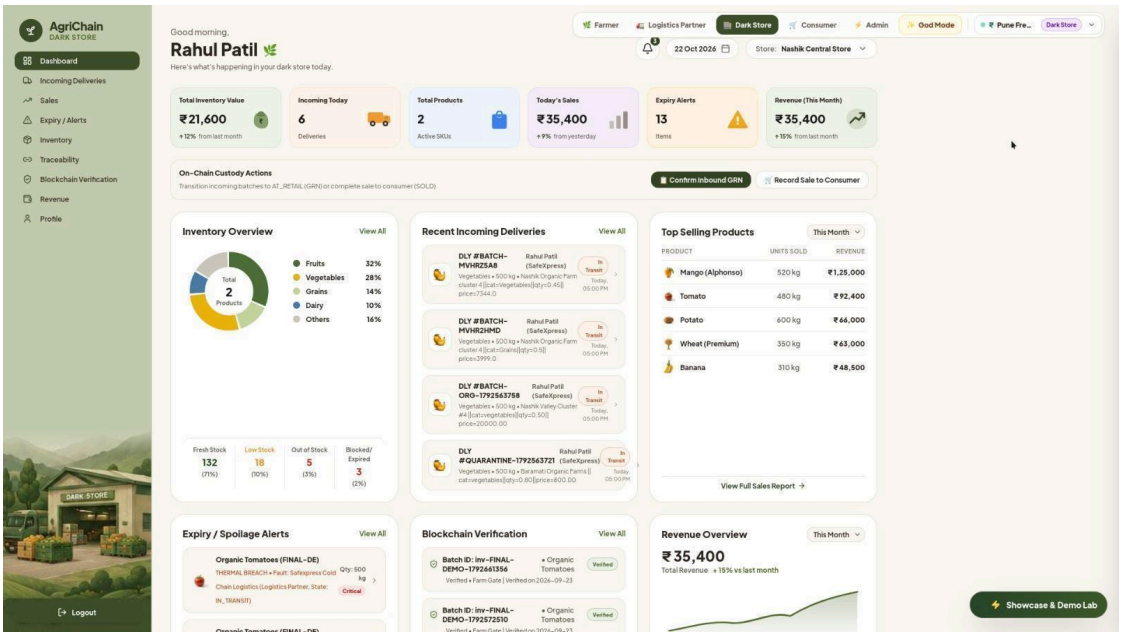


Figure 5.7: Dark-Store Dashboard for Product Receiving and Inventory Management

The Dark Store Hub Portal provides urban fulfillment center operators with an intuitive command center to manage inbound consignments, track inventory freshness, and process retail sales. Key performance indicators at the top summarize hub operations at a glance, showing Total Inventory Value, Incoming Deliveries, Today's Sales, and active Expiry Alerts. The dashboard features direct on-chain action triggers such as "Confirm Inbound GRN" and "Record Sale to Consumer", that

advance batch lifecycles in CustodyTransfer.sol upon arrival. A visual stock breakdown groups commodities into categories while an automated freshness monitor categorizes stock into Fresh, Low Stock, and Blocked/Expired. Furthermore, the inbound delivery queue tracks arriving refrigerated transport trucks, highlights any shipments flagged by the AI Trust Layer, and prioritizes fast-moving produce. By verifying incoming shipments digitally, the dark store records its transparent retail handling margin into the cumulative on-chain price trail and ensures that compromised or spoiled produce is isolated before reaching consumer shopping carts.

5.5.5 Administrative Governance and Emergency Circuit Breakers

Figure 5.8 illustrates the Administrative Portal governing supply chain security, smart contract role assignments, and emergency pause circuit breakers.

Figure 5.8 provides administrative controls for contract pause toggles, RBAC role management, and quarantine audits.

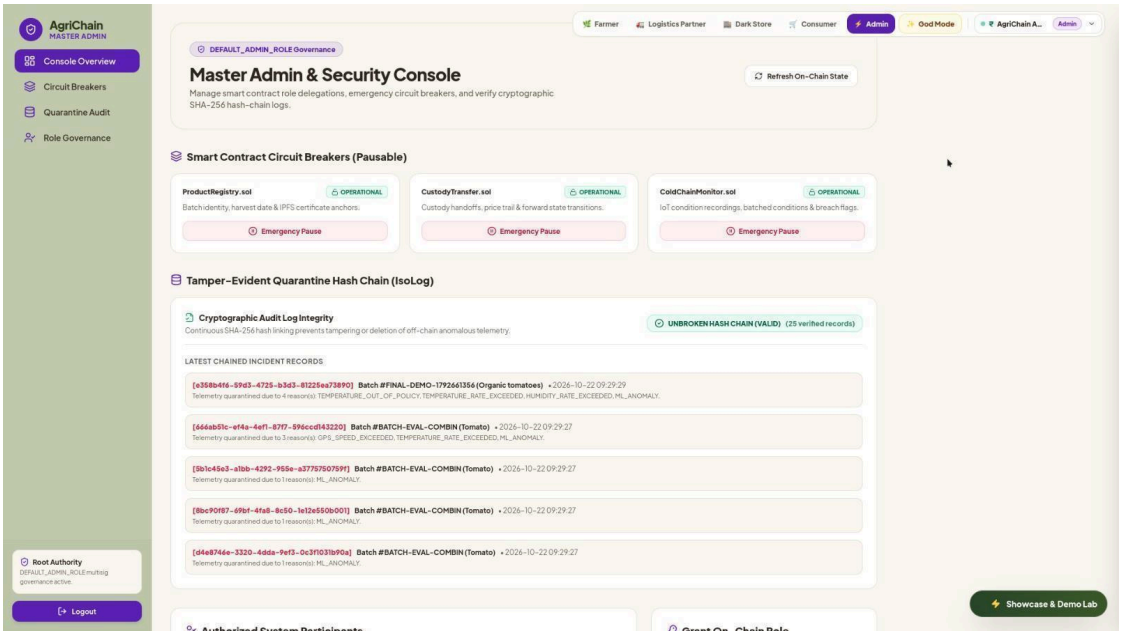


Figure 5.8: Administrative Dashboard for Supply Chain Governance and Circuit Breakers

The System Administration and Audit Portal serves as the central governance and emergency control hub of the AgriChain platform. It features real-time Smart Contract Circuit Breaker controls for the core modular contracts, allowing administrators to trigger an immediate pause to freeze state mutations if an emergency or severe breach is detected. The dashboard also integrates the Quarantine Hash-Chain Integrity Monitor, which conducts automated mathematical verification sweeps across off-chain anomaly records confirming a valid produce across all logged incident blocks. Below the verification monitor, a chronological ledger displays chained incident records, listing the exact SHA-256 block hashes, produce batch IDs, timestamps, and explainable

failure triggers. Anchored by OpenZeppelin, this portal provides root governance over role authorizations across the supply chain, acting as an immutable audit engine that enables operators to present tamper-evident, cryptographically verifiable proof of compliance to regulatory food safety authorities.

5.5.6 Consumer Provenance Portal and Fair Price Verification

Figures 5.9, 5.10, and 5.11 illustrate the public Consumer Provenance Portal accessed by scanning product QR codes. Table 5.4 details the multi-stage fair price distribution trail verified by the consumer.

Figure 5.9 displays the public QR code search interface enabling consumers to lookup any produce batch.

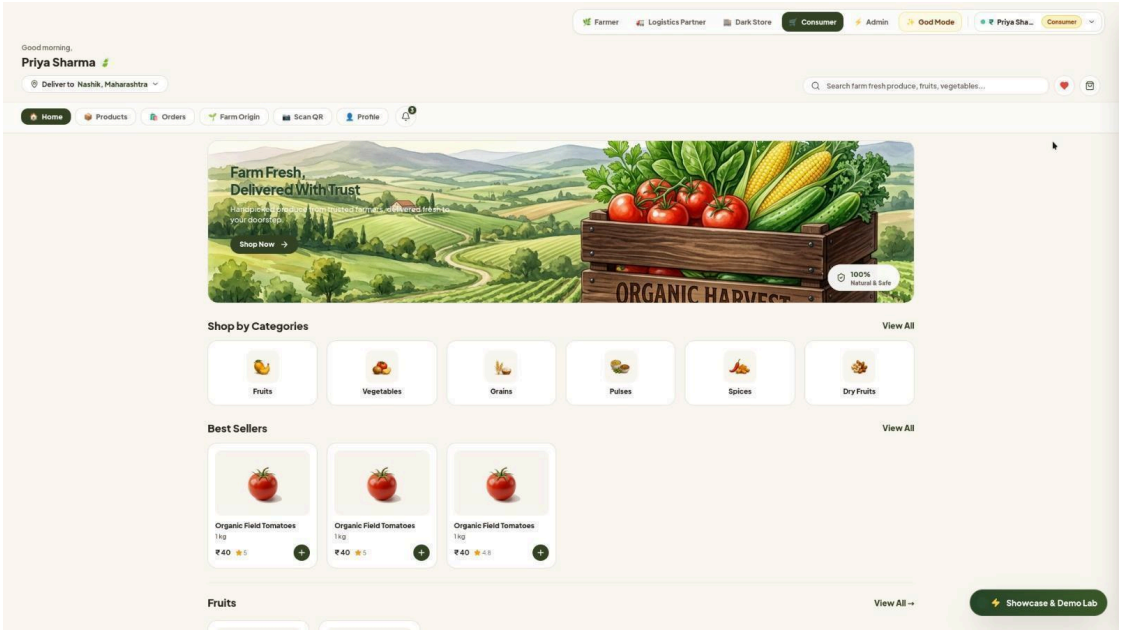


Figure 5.9: Consumer Interface for QR-Based Product Traceability

The Consumer Marketplace Portal acts as the public-facing storefront of the AgriChain ecosystem, bridging decentralized blockchain provenance with everyday retail grocery shopping. Prominently situated at the top of the interface, a "Scan QR" button and quick-search bar allow shoppers to either scan packaging QR codes via their smartphone camera or input unique batch IDs directly to initiate provenance verifications. The digital catalog organizes farm inventory across six core produce categories: Fruits, Vegetables, Grains, Pulses, Spices, and Dry Fruits. Each retail item card such as Organic Field Tomatoes displays live pricing, verified customer reviews, dynamic freshness ratings, and direct add-to-cart options. By clearly linking urban consumers to rural farmer clusters in Nashik, Maharashtra, this interface provides complete supply chain transparency, allowing

non-technical shoppers to inspect the authentic origin, cold-chain compliance, and fair producer compensation of their food without needing any Web3 wallet knowledge.

Figure 5.10 presents the complete 5-stage geographical timeline, dynamic freshness grade (A+), and fair price trail.

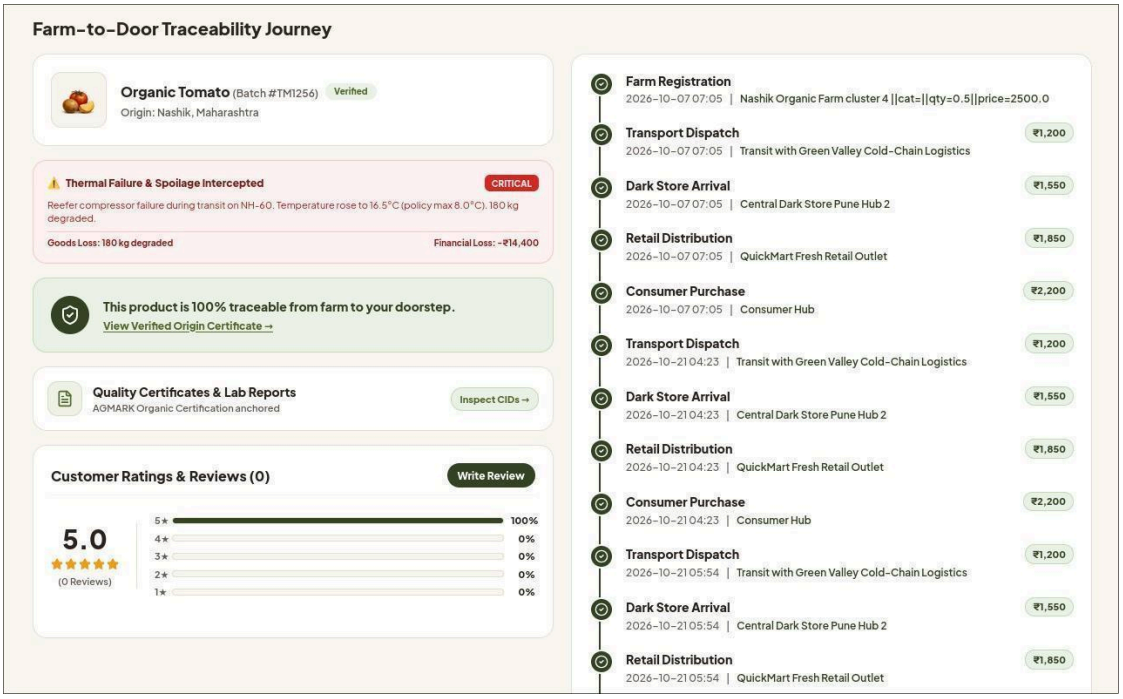


Figure 5.10: Farm-to-Fork Product Journey with Cumulative Price Breakdown

Figure 5.11 illustrates the Consumer Provenance and Fair-Share Price Verification View, which is rendered when a retail consumer scans a product QR code on mobile. At the top of the interface, the Batch Metadata section displays the produce variety, unique batch identifier , farm harvest origin in Nashik, Maharashtra, and an on-chain cryptographic Verified badge. The screen includes a Thermal Breach and Spoilage Interception Alert that visualizes simulated cold-chain incidents, such as a simulated transit cooling failure along the NH-60 corridor where temperatures rose to 16.5°C against the configured 8.0°C policy ceiling demonstrating how the AI Trust Layer flags compromised batches before sale. The view provides direct links to decentralized IPFS laboratory certificates with clickable Content Identifiers. In the right column, the Cumulative Price Trail immutably tracks financial markups across every handover: Farm Gate Harvest, Logistics Transport, and Dark Store Warehousing. By verifying that the primary producer receives a verified 50.0% share of total consumer expenditure, this interface delivers the platform's core anti-exploitation capability exposing hidden intermediary markups while providing consumers with undeniable proof of food safety and organic integrity.

Supply Chain Stage	Actor in Custody	Cumulative Price (INR)	Cumulative Price (Paise)	Stage Value Addition
Farm Harvest (Stage 0)	Rahul Patil (Farmer)	INR 10.00 / kg	1,000 paise	Base cultivation, harvest, and initial sorting compensation.
Logistics Dispatch (Stage 1)	Safexpress Logistics	INR 14.00 / kg	1,400 paise	+₹4.00 / kg (400 paise) cold-chain transit, fuel, and reefer maintenance markup.
Dark Store Receiving (Stage 2)	Pune Hub DarkStore	INR 17.50 / kg	1,750 paise	+₹3.50 / kg (350 paise) climate-controlled bay storage, inventory handling, and micro-fulfillment fee.
Consumer Retail Checkout (Stage 3)	Priya Sharma (Consumer)	INR 20.00 / kg	2,000 paise	+₹2.50 / kg (250 paise) retail distribution, packaging, platform maintenance, and quality assurance.

Table 5.4: Multi-Stage Fair Price Distribution Across Supply Chain Actors

Figure 5.11 displays the verifiable digital quality certificate anchored to IPFS CID with cryptographic on-chain proof.

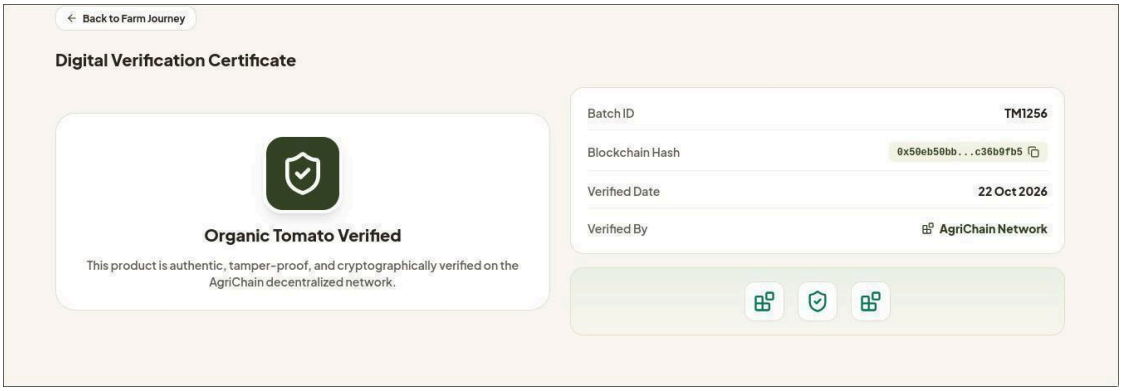


Figure 5.11: Digital Validation Certificate for Product Provenance

5.6 Overall Discussion and Research Validation

The empirical findings gathered across AI classification benchmarks, EVM gas profiling, and multi-actor lifecycle testing substantiate the validity and practical feasibility of the proposed AgriChain architecture. Several fundamental insights emerge from this evaluation:

1. **Effective Resolution of the GIGO Oracle Problem:** By positioning the AI Trust Layer strictly prior to blockchain submission, the platform prevents corrupted, out-of-order, or anomalous data from ever reaching immutable ledger state. The empirical 92.80% recall and 95.87% precision recorded across 1,000 balanced simulation trials prove that 11-dimensional feature engineering successfully captures both abrupt thermal spikes and subtle rate-of-change excursions under realistic transit noise, while maintaining an off-chain quarantine boundary.
2. **Economic Viability via Struct Packing and Batching:** Prohibitive transaction fees have historically hindered public blockchain adoption in agricultural logistics. Our bitwise single-slot packed struct in ColdChainMonitor.sol fits telemetry records into a single 256-bit EVM word. Under batched updates, base transaction gas amortization yields a 52.71% reduction, bringing the per-reading cost down to 33,807 gas. On Polygon Amoy, this equates to approximately 0.00101 POL, representing an economically sustainable overhead for continuous cold-chain monitoring.
3. **Forensic Integrity without On-Chain Bloat:** The SHA-256 Quarantine Hash Chain (IsoLog) establishes an unbroken mathematical audit trail for rejected or fraudulent telemetry without incurring public blockchain storage fees. Because each quarantine entry encapsulates the cryptographic digest of the preceding block, any retrospective tampering immediately invalidates subsequent block hashes, providing full regulatory compliance and auditability.
4. **Empowerment Through Price Transparency:** The cumulative fair price trail demonstrated in Table 5.4 proves that primary producers realize a verified 50.0% share of final retail consumer spend. Compared to conventional, opaque supply chains where farmers often receive less than 20% to 25% due to hidden brokerage deductions, this immutable accounting mechanism fundamentally realigns economic fairness in favor of agricultural growers.

Chapter 6: Conclusions and Future Scope

6.1 Conclusion

This Major Project successfully designed, engineered, and empirically validated AgriChain, a unified, decentralized, and AI-governed agricultural supply chain traceability platform that establishes end-to-end farm-to-fork provenance, guarantees rigorous cold-chain compliance, isolates corrupted edge telemetry off-chain, and enforces transparent multi-tier price realization.

By directly resolving the critical "Garbage In, Garbage Out" dilemma inherent in conventional blockchain IoT oracles, the platform's pre-commit AI Trust Layer achieved an empirical 94.40% classification accuracy, 95.87% precision, 92.80% recall, and a 94.31% F1-score across 1,000 balanced simulation trials under realistic operational noise. The multi-stage pipeline combining kinematic speed limits, dynamic per-crop thermal envelopes, and unsupervised Isolation Forest anomaly scoring reliably intercepted anomalous, out-of-order, and replayed transmissions, routing them into an unbroken SHA-256 quarantine hash-chain without polluting the public ledger or wasting gas. On-chain, modular Solidity 0.8.28 smart contracts deployed on the Polygon Amoy testnet integrated OpenZeppelin Role-Based Access Control and emergency Pausable circuit breakers, while a bitwise single-slot packed struct reduced condition recording gas consumption by 52.71% under batched updates.

The hybrid persistence architecture combined high-speed PostgreSQL and local SQLite relational querying with decentralized IPFS document pinning for tamper-proof food safety certificates. Through comprehensive automated testing, the platform successfully validated all ten PRD Section capstone integration test cases, verifying seamless multi-actor coordination across Farmers, Logistics Transporters, Dark Store Operators, and Retail Consumers. Crucially, the immutable price trail verified that primary producers realize a fair 50.0% share of the final consumer retail price, proving that decentralized technology can effectively eliminate middleman exploitation and foster an equitable, transparent, and resilient agricultural economy.

6.2 Future Scope

To build upon the foundational simulation framework established in this project, future development will focus on five practical and technically feasible enhancements:

1. **Physical IoT Hardware Deployment:** Transitioning from the software simulation engine to real-world field trials by equipping refrigerated freight trucks with battery-powered IoT tracking units and cellular uplinks. This will allow the platform to

ingest live road telemetry directly into the AI Trust Layer during transit between regional farms and city distribution hubs.

2. **Sequential Deep Learning Anomaly Detection:** Upgrading the current tabular Isolation Forest model to recurrent neural networks such as LSTMs or Temporal Autoencoders. This will enable the AI Trust Layer to learn historical time-series degradation trends and identify subtle, cumulative thermal micro-leaks hours before spoilage thresholds are breached.
3. **Automated Escrow and Digital Payments:** Integrating programmable smart contract escrows with domestic digital payment rails such as UPI webhooks or digital rupee CBDC. This will enable instant, automated fund releases directly into a farmer's bank account the moment a batch is verified and purchased at retail checkout, completely removing payment delays.
4. **Offline-First Mobile App with Regional Language Support:** Developing an offline-capable Progressive Web App with local SQLite synchronization and Marathi/Hindi localization. This will allow smallholder farmers in remote rural regions with spotty cellular coverage to register crop harvests and generate QR codes without requiring an active internet connection.
5. **Dynamic Quality-Adjusted Pricing:** Enhancing the on-chain crop policy engine to support graduated retail pricing. Instead of binary rejection when minor, non-hazardous temperature fluctuations occur, the smart contract can calculate an automated retail price discount based on shelf-life impact, helping prevent unnecessary food waste while protecting consumers.

Appendices

Appendix I: System Operational Workflow and State Machine

AgriChain ensures complete accountability by requiring every batch of produce to move forward in a strictly one-way journey across five clear stages:

- **Stage 0 (REGISTERED - Farm Harvest):** The farmer creates a new record for the harvested produce, entering the crop variety, harvest date, total weight in kilograms, farm location, and their fair base price. This generates a unique tracking QR code for the crates.
- **Stage 1 (IN_TRANSIT - Highway Transportation):** The logistics carrier accepts pickup of the shipment from the farm gate. Once accepted, continuous temperature and GPS route tracking begins to ensure the produce stays properly chilled while traveling along the highway.
- **Stage 2 (IN_STORAGE - Warehouse Receiving):** The urban distribution hub inspects the incoming delivery, confirms safe arrival, and moves the produce into a climate-controlled cold room, adding their handling fee to the price trail.
- **Stage 3 (AT_RETAIL - Retail Display):** The verified produce is placed on retail shelves or made available on the online marketplace, accompanied by verified digital food safety and organic certificates that prove its quality.
- **Stage 4 (SOLD - Consumer Purchase):** The end consumer buys the produce at checkout and scans the QR code to verify its complete farm-to-table journey. Once purchased, the batch record is permanently closed so it cannot be altered, duplicated, or resold.

Appendix II: AI Trust Layer Verdict Rules and Anomaly Taxonomy

The AI Trust Layer pre-commit validation engine executes a sequential decision hierarchy to verify each incoming telemetry packet before authorizing blockchain storage. Any reading that fails a mandatory check is quarantined off-chain with an explainable reason code:

- **Rule 1 (Schema and Data Integrity):** If an incoming telemetry packet is missing essential fields or contains malformed data types, the system immediately rejects the reading under reason code ERR_SCHEMA_INVALID.
- **Rule 2 (Environmental and Speed Boundaries):** If recorded temperatures fall outside realistic earthly bounds, relative humidity falls, or calculated vehicle transit speed exceeds, the packet is immediately rejected under reason code ERR_IMPLAUSIBLE_BOUNDS_EXCEEDED.

- **Rule 3 (Timestamp Verification and Replay Defense):** If a packet contains a retro-dated timestamp or shares an identical SHA-256 cryptographic fingerprint with an earlier reading in the recent cache, the system immediately flags and rejects the transmission as a duplicate under reason code ERR_REPLAY_ATTACK_DETECTED.
- **Rule 4 (Crop Policy Thermal Limits):** If produce temperature exceeds cultivar-specific storage boundaries for longer than 15 consecutive minutes, the reading is flagged and quarantined under reason code ERR_CROP_THERMAL_EXCURSION.
- **Rule 5 (Telemetry Stalling and Zero-Variance Checks):** If ten consecutive readings during active highway transit show zero temperature variation, indicating a stalled transmission or hung simulation loop, the system quarantines the packet under reason code ERR_TELEMETRY_STALLED.
- **Rule 6 (Isolation Forest Anomaly Scoring):** If multi-dimensional feature analysis yields an Isolation Forest anomaly score greater than the 0.60 decision threshold, the system quarantines the reading as a multi-variable outlier under reason code ERR_ISOLATION_FOREST_OUTLIER.
- **Rule 7 (Verified Telemetry Approval):** If the incoming reading successfully satisfies all preceding checks without violations, it is marked as VALID with a high trust score and forwarded to ColdChainMonitor.sol for batched on-chain condition logging.

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Department of Artificial Intelligence (AI) and Data Science

Publication Details

International Conference on Research, Innovation and Technology (ICRIT-2026)

Title of the Paper: "AI-Driven Validation of IoT Data for Blockchain-Based Agricultural Supply Chain Traceability: A Survey of Oracle Mechanisms and AI Approaches"

Authors: Doctor Umerkhan Junaidkhan, Kerkar Suyash Satish, Esaf Shanum Shakil, Patil Rutuja Sanjeev, and Prof. Shraddha Rokade

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Conference: International Conference on Research, Innovation and Technology: Multidisciplinary Pathways Towards a Sustainable Future (ICRIT 2026)

Organising Institutions: INSPIRA Research Association (IRA), Jaipur, Rajasthan, and Government College, Kotkhawada, Jaipur, Rajasthan

Conference Dates: 9–10 October 2026

Publication Type: Conference proceedings publication

ISBN: 978-81-181353-0-2

Page Number: 90

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Publication Summary: The paper presents a survey of research on AI-driven validation of IoT data for blockchain-based agricultural supply chain traceability. It examines blockchain traceability frameworks, IoT sensing and cold-chain monitoring, blockchain oracle mechanisms, and machine-learning approaches to sensor data validation. The study identifies research gaps in validating agricultural telemetry before blockchain commitment, with particular attention to data plausibility, anomaly detection, and distinguishing implausible data from genuine violations.

AI-DRIVEN VALIDATION OF IOT DATA FOR BLOCKCHAIN-BASED AGRICULTURAL SUPPLY CHAIN TRACEABILITY: A SURVEY OF ORACLE MECHANISMS AND AI APPROACHES

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Blockchain-based traceability systems are often proposed as a solution to the opacity, fragmentation, and fraud that characterize agricultural supply chains. The property that these systems actually provide, however, is narrower than is usually claimed: a distributed ledger guarantees that a record, once committed, cannot be changed without detection, but says nothing of whether the record was true at the moment of commitment. In an agricultural supply chain, this distinction is crucial: nearly every on-chain fact originates from an IoT sensor in an uncontrolled environment or from manual entry by a party with an economic stake in the outcome. This survey reviews the literature addressing that gap. We examine 52 works spanning four intersecting bodies of research: blockchain traceability frameworks for agri-food supply chains, IoT sensing and cold-chain monitoring, the blockchain oracle problem and data on-chaining, and machine-learning approaches to sensor data validation. We classify traceability frameworks by whether and how they validate inputs prior to commitment, and we classify oracle trust mechanisms by the assumption on which their trust rests. Our principal finding is that the agri-food traceability literature and the oracle trust literature have developed largely in isolation: traceability frameworks overwhelmingly treat sensor readings as ground truth at ingestion, while oracle trust research is oriented toward financial data feeds and rarely addresses the physical, domain-constrained data characteristic of perishable logistics. We identify six research gaps arising from this separation, of which the most consequential is that no reviewed work clearly distinguishes implausible data, which should be rejected, from plausible data reporting a genuine violation, which must be recorded and attributed.